



Three-dimensional topology optimization of thermal-fluid-structural problems for cooling system design

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Abstract

In the present study, a topology optimization method of thermal-fluid-structural problems is researched to design the three-dimensional heat sink with load-carrying capability. The optimization is formulated as a mean temperature minimization problem controlled by Navier-Stokes (N-S) equations as well as energy balance and linear elasticity equations. In order to prevent an unrealistic and low load-carrying design, the power dissipation of the fluid device and the normal displacement on the load-carrying surface are taken as constraints. A parallel solver of multi-physics topology optimization problems is built-in Open Field Operation And Manipulation (OpenFOAM) software. The continuous adjoint method is adopted for the sensitivity analysis to make the best use of built-in solvers. With the developed tool, the three-dimensional (3D) thermal-fluid topology optimization is studied. It is found that the Darcy number, which is suitable for fluid design, may cause severe problems in thermal-fluid optimization. The structural features of 3D thermal-fluid-structural problems are also investigated. The “2D extruded designs” are helpful to improve the structural stiffness, and channels with a larger aspect ratio in high-temperature areas improve the cooling performance.

Keywords Topology optimization · Thermal-fluid-structural optimization problem · Continuous adjoint method · Parallel computing · OpenFOAM

1 Introduction

Cooling systems are widely used in engineering products and processes, such as batteries, chips, and injection molding. With the development of manufacturing technology, the liquid-cooled heat sink holds great potential as a robust and effective cooling device for removing high heat flux. However, the

inappropriate design of cooling channels may reduce the efficiency of the systems and increase the probability of product failure.

Some researchers (Au et al. 2011; Li 2001; Park and Pham 2009; Xu et al. 2010; Yu et al. 2011) applied the heuristic method to design high-performance heat sink. The optimization process is very efficient because no simulation is required; however, the correctness of the obtained results cannot be guaranteed. Another widely used approach is the simulation-based parametric method (Dang and Park 2011; Li et al. 2012; Lin 2002; Qiao 2006; Wang et al. 2016). The merit of this method is that simulations can be handled in commercial software. However, this method is limited to very low design freedoms, and final results strongly depend on initial configurations.

In topology optimization, an innovative structure or product is designed by distributing materials in an optimal way. It has originated from structural optimization problems (Bendsoe and Kikuchi 1988; Sigmund 1997; Sigmund 2001) and is now extended to multiple-physics problems, such as acoustics, fluidics, optics, and thermal (Duhring et al. 2008; Gersborg-Hansen et al. 2006; Nomura et al. 2007; Sigmund and Hougaard 2008). The topology optimization of fluid flow systems aims at creating new

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fluid-solid boundaries so that objectives like pressure drop or power dissipation can be minimized. Borrvall and Petersson (2003) studied fluid topology optimization problems by the density-based topology optimization method. The design domain was considered a porous material whose permeability was determined by the pseudo-density. By varying the local permeability, solid and fluid regions were smoothly interpolated. Similar models have been applied to different flow problems, such as Darcy-Stokes flows (Guest and Prevost 2006; Wiker et al. 2007), Navier-Stokes (N-S) flows with low and moderate Reynolds numbers (Deng et al. 2010; Gersborg-Hansen et al. 2005; Olesen et al. 2006), slightly compressible N-S flows (Evgrafov 2006), unsteady N-S flows (Deng et al. 2011; Kreissl et al. 2011), and turbulent flows (Dilgen et al. 2018b; Othmer et al. 2007; Yoon 2016). Furthermore, optimized results have been used to investigate thermal-hydraulic performances experimentally (Li et al. 2019a, b).

Many researchers studied multiple-physics-coupled topology optimization problems. Yoon (2010) designed a 2D heat dissipating structure that utilized forced convective heat transfer. The main concern of this work was to minimize thermal compliance. A weak coupling that ignored buoyancy and viscous dissipation inside the fluid was adopted. A friction force was added to N-S equations. Material properties, such as specific heat capacity, conductivity, and density, were interpolated with respect to the pseudo-density field. Kontoleon et al. (2013) adopted the Spalart-Allmaras model to analyze turbulent flows in a cooling system. The exact continuous adjoint state system was derived without using frozen turbulence assumption. Thermal-fluid problems have been also studied by different topology optimization methods, such as level set (Yaji et al. 2015) and component-based method (Yu et al. 2019). Yoon (2012) first studied a more complicated topology optimization problem coupled with electrical, fluid, thermal, and structural parameters and developed a novel rigorous monolithic analysis and optimization framework for micro-actuators. Zhao et al. (2019) designed a layout of straight cooling pipes on the cross-section considering heat transfer and carrying load and presented a multi-phase topology optimization approach consisting of solid, liquid, and void.

For topology optimization problems, more design variables lead to more detailed designs and simultaneously generate a greater computational costs. Therefore, the parallel computing technique has attracted great attention in this area. Aage et al. (2017) studied the topology optimization of the internal structure of a full-scale aeroplane wing with giga-voxel resolution. For the topology optimization of flow problems, Yaji et al. (2018) studied unsteady thermal-fluid problems with about 51 million design variables. Both state and adjoint problems were solved by hybrid MPI/OpenMP parallel programming. Some researchers have also employed parallel computing based on the PETSc framework. Alexandersen et al. (2016) studied the design of natural convection cooling heat sinks with 20–330 million

degrees of freedom. Dilgen et al. (2018b) studied turbulent flow heat transfer problems with 216,000 design variables.

The calculation of accurate sensitivities is one of the main challenges in multi-physics topology optimization. The adjoint method is the most popular approach because its necessary derivatives are independent of the number of the design variables. For fluid dynamics problems, complicated functions and segregated solvers are usually adopted, making it an error-prone task to derive the adjoint system manually. In order to avoid this problem, the discrete adjoint method has implemented based on the automatic differentiation (AD) approach (Dilgen et al. 2018a, b; Farrell et al. 2013; Marta et al. 2007; Nørgaard et al. 2017). Some software, such as FEniCS and COMSOL Multiphysics, provide AD in a black-box manner; however, it usually results in immense memory requirements to store intermediate variables in the reversed propagation of derivatives (Papoutsis-Kiachagias and Giannakoglou 2014). Instead of using AD in a black-box manner, Marta et al. (2007) proposed an “ADjoint” approach to compute certain terms of discrete adjoint equations. Dilgen et al. (2018a, b) applied this method to turbulent flow topology optimization problems to reduce the memory requirement and compute exact sensitivities. The continuous adjoint method is usually simpler than the discrete adjoint method in terms of implementation and requires less memory (Giles and Pierce 2000; Nadarajah and Jameson 2013; Peter and Dwight 2010). Othmer et al. (2007) derived a continuous adjoint formulation for fluid topology optimization problems and implemented it in OpenFOAM.

Previous studies of fluid-thermal topology optimization mainly focused on improving the cooling performance; however, impractical results may be obtained in the three-dimensional design, such as the suspending solid structure. Furthermore, the existence of channels could reduce the load-carrying capability, which is also important during service. For example, an injection mold needs enough stiffness to resist the high injection pressure and clamping force.

The choice of Darcy number in thermal-fluid problems is another issue that needs to be addressed. Due to the nature of the penalization, the Darcy number should be large enough to enforce the velocity negligible in solid areas. Olesen et al. (2006) studied 2D fluid flow problems and suggested that the Darcy number should be less than 10^{-5} to make the solid material impermeable. Most of the researchers adopted 10^{-5} and a lower Darcy number of 10^{-6} for 3D thermal-fluid problems is used by Dilgen et al. (2018b). However, the choice of Darcy number for thermal-fluid problems has not been investigated. In this work, an interesting phenomenon can be observed that even if the Darcy number is low enough to punish the velocity close to 0, it cannot eliminate the thermal convection term in the solid area.

In the present work, the built-in solver in OpenFOAM (names `ajointShapeOptimizationFoam`) is extended to perform topology optimization of thermal-fluid-structural problems. The Brinkman penalization method is adopted to build unified governing

equations that model both fluid and solid areas. With the developed tool, the choice of Darcy number for 3D thermal-fluid problems is investigated first, then the influences of the displacement constraint and thermal expansion on structural features are discussed in detail. The remaining paper is organized as follows. In Section 2, the details of the topology optimization process and how to build solvers in OpenFOAM are introduced. In Section 3.1, the accuracy of the adjoint solver is verified first, and 2D heat sinks with different power dissipation are studied. In Section 3.2, the choice of the Darcy number for thermal-fluid problems is discussed. In addition, the influences of thermal expansion and external force on a 3D heat sink are studied. Finally, conclusions are presented in Section 4.

2 Density-based topology optimization for thermal-fluid-structural problems

2.1 Governing equations

The present work focuses on designing a high-performance cooling system at moderate Reynolds numbers by considering the deformations caused by thermal expansion and loading force. The problem is modeled by steady-state incompressible N-S equations as well as energy balance and linear elasticity equations. The assumptions for solving this problem are presented below.

- All material properties are independent of temperature.
- Heat caused by viscous dissipation is neglected.
- Impacts of fluid flow on structural deformation are neglected.
- Effects of mechanical deformations on the configuration of the fluid region are neglected.

Therefore, the governing equations of the model are presented below, and the schematic diagram of the analysis domain including the definitions of boundaries is displayed in Fig. 1.

N-S equations:

$$\begin{cases} -\nabla \cdot \mathbf{u} = 0 \\ \rho(\mathbf{u} \cdot \nabla) \mathbf{u} - \eta \nabla \cdot (\nabla \mathbf{u} + \nabla \mathbf{u}^T) + \nabla p - \mathbf{F}_b = 0 \\ \mathbf{u} = \mathbf{u}_D \\ [\eta(\nabla \mathbf{u} + \nabla \mathbf{u}^T) - p \mathbf{I}] \cdot \mathbf{n} = \mathbf{g}_N \end{cases} \quad \begin{array}{l} \text{in } \Omega \\ \text{in } \Omega \\ \text{on } \Gamma_D^F \\ \text{on } \Gamma_N^F \end{array} \quad (1)$$

Energy conservation equation:

$$\begin{cases} \rho c(\mathbf{u} \cdot \nabla T) = \nabla \cdot (k \nabla T) + Q \\ T = T_D \\ k \nabla T \cdot \mathbf{n} = q_N \end{cases} \quad \begin{array}{l} \text{in } \Omega \\ \text{on } \Gamma_D^T \\ \text{on } \Gamma_N^T \end{array} \quad (2)$$

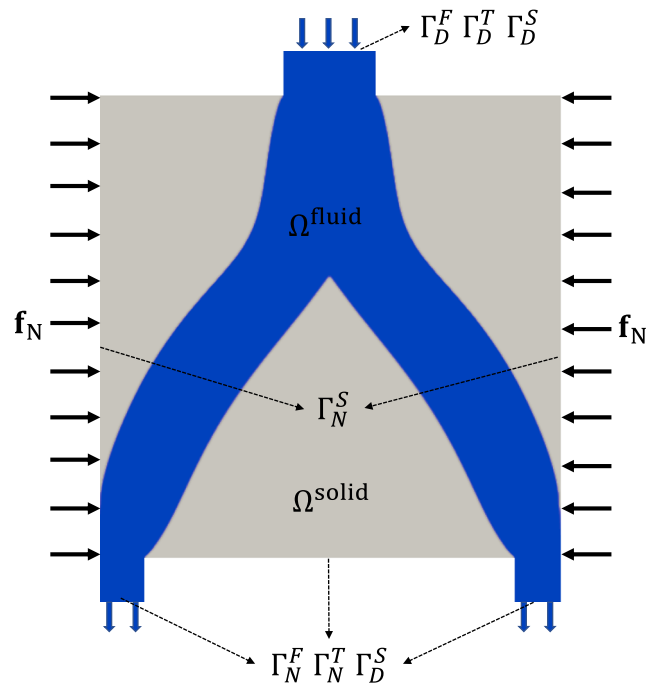


Fig. 1 Schematic diagram of the analysis domain including the definitions of boundaries

Linear elasticity equation:

$$\begin{cases} \nabla \cdot [\mu(\nabla \mathbf{D} + \nabla \mathbf{D}^T) + \lambda(\nabla \cdot \mathbf{D}) \mathbf{I} - \alpha_T(3\lambda + 2\mu)(T - T_{ref}) \mathbf{I}] = 0 & \text{in } \Omega \\ \mathbf{D} = \mathbf{D}_D & \text{on } \Gamma_D^S \\ [\mu(\nabla \mathbf{D} + \nabla \mathbf{D}^T) + \lambda(\nabla \cdot \mathbf{D}) \mathbf{I} - \alpha_T(3\lambda + 2\mu)(T - T_{ref}) \mathbf{I}] \cdot \mathbf{n} = \mathbf{f}_N & \text{on } \Gamma_N^S \end{cases} \quad (3)$$

where Ω denotes the computational domain and Γ is the boundary, the subscripts D and N represent the Dirichlet part and the Neumann part of the boundary, the superscripts F , T , and S represent the fluid, thermal, and solid mechanics problems, respectively. ρ is the fluid mass density, and $\mathbf{u}$ and p are the velocity and pressure fields, respectively. η is the dynamic viscosity, $\mathbf{F}_b$ is the body force of the fluid. $\mathbf{u}_D$ and $\mathbf{g}_N$ are the specified velocity and stress distributions on the boundaries Γ_D^F and Γ_N^F , respectively. T is the temperature field, $\mathbf{n}$ is the unitary outward normal, c is the specific heat capacity, k is the thermal conductivity, Q is the heat source term, and T_D and q_N are the specified temperature and heat flux on the boundaries Γ_D^T and Γ_N^T , respectively. $\mathbf{D}$ is the solid displacement field, μ and λ are Lamé parameters for the solid, T_{ref} is the reference temperature and was set to 0 in the present analysis, α_T is the thermal expansion coefficient, $\mathbf{I}$ is the second-order unit tensor, and $\mathbf{D}_D$ and $\mathbf{f}_N$ are the specified displacement and stress distribution on the boundaries Γ_D^S and Γ_N^S , respectively.

2.2 Interpolation of material properties

The density-based topology optimization method is adopted in the present study. The concept of pseudo-density field (γ) in the

$[0, 1]$ domain is used to describe the distribution of materials. $\gamma = 0$ represents the solid, and $\gamma = 1$ represents the fluid. According to the work of Borrvall and Petersson (2003), a friction force proportional to the fluid velocity is introduced into N-S equations:

$$\mathbf{F}_b = -\alpha_f \mathbf{u} \quad (4)$$

where α_f is the inverse of the local permeability. In the fluid region, α_f is equal to 0, and it signifies that no artificial friction force is added. In the solid region, α_f is equal to the relatively large number $\alpha_{\max}$ to set the velocity to near 0. α_f is related to the pseudo-density field by the following interpolation:

$$\alpha_f = \alpha_{\max} \frac{q_\alpha(1-\gamma)}{q_\alpha + \gamma} \quad (5)$$

where q_α is a positive parameter for adjusting the shape of α_f as shown in Fig. 2.

For the energy conservation equation, the thermal conductivity is interpolated from the conductivity of the solid (k_s) to that of the fluid (k_f):

$$k = k_f + (k_s - k_f) \frac{q_k(1-\gamma)}{q_k + \gamma} \quad (6)$$

where q_k is a positive parameter. In the current study, the mass density (ρ) and the specific heat capacity (c) are not interpolated. The properties of the fluid are used because the convection term yielded to 0 in the solid region.

For the linear elasticity equation, Lamé's parameters are related to Young's modulus (E) and Poisson's ratio (ν) by the following relationships:

$$\mu = \frac{E}{2(1+\nu)}, \quad \lambda = \frac{E\nu}{(1-2\nu)(1+\nu)} \quad (7)$$

Although the finite element method (FEM) is the most popular way for solving structural topology optimization

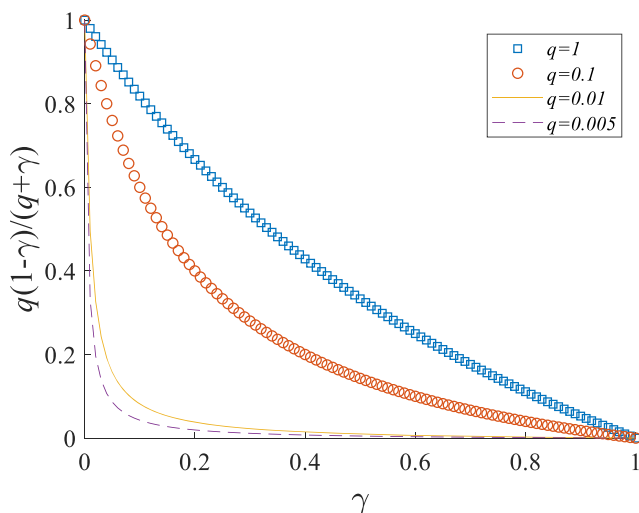


Fig. 2 Interpolation function for different values of the curvature parameter q

problems, considering OpenFOAM based on the finite volume method (FVM), the built-in FVM solver is adopted in the present study. The material interpolation model in FEM (elastic modulus is interpolated from solid ($E_{\max}$) to void ($E_{\min}$)) is also used:

$$E = E_{\min} + (E_{\max} - E_{\min}) \frac{q_E(1-\gamma)}{q_E + \gamma} \quad (8)$$

where q_E is a positive parameter and $E_{\min} = 0.0001E_{\max}$.

2.3 Topology optimization formulation

Cooling capacity is the main concern for designing a cooling system, and it can be measured by the mean temperature:

$$\Psi = \frac{1}{|\Omega|} \int_{\Omega} T d\Omega \quad (9)$$

where Ψ denotes the objective function, and $|\Omega|$ denotes the volume of the computational domain.

In order to prevent an unrealistic and low load-carrying design, the power dissipation of the fluid device and the normal displacement on the load-carrying surface are considered constraints:

$$J = - \int_{\Gamma_{\text{in,out}}} \left(p + \frac{1}{2} \mathbf{u} \cdot \mathbf{u} \right) \mathbf{u} \cdot \mathbf{n} d\Gamma \quad (10)$$

$$S = \int_{\Gamma_{\text{force}}} (\mathbf{D} \cdot \mathbf{n})^2 d\Gamma \quad (11)$$

where J denotes the power dissipation, and S is the normal displacement.

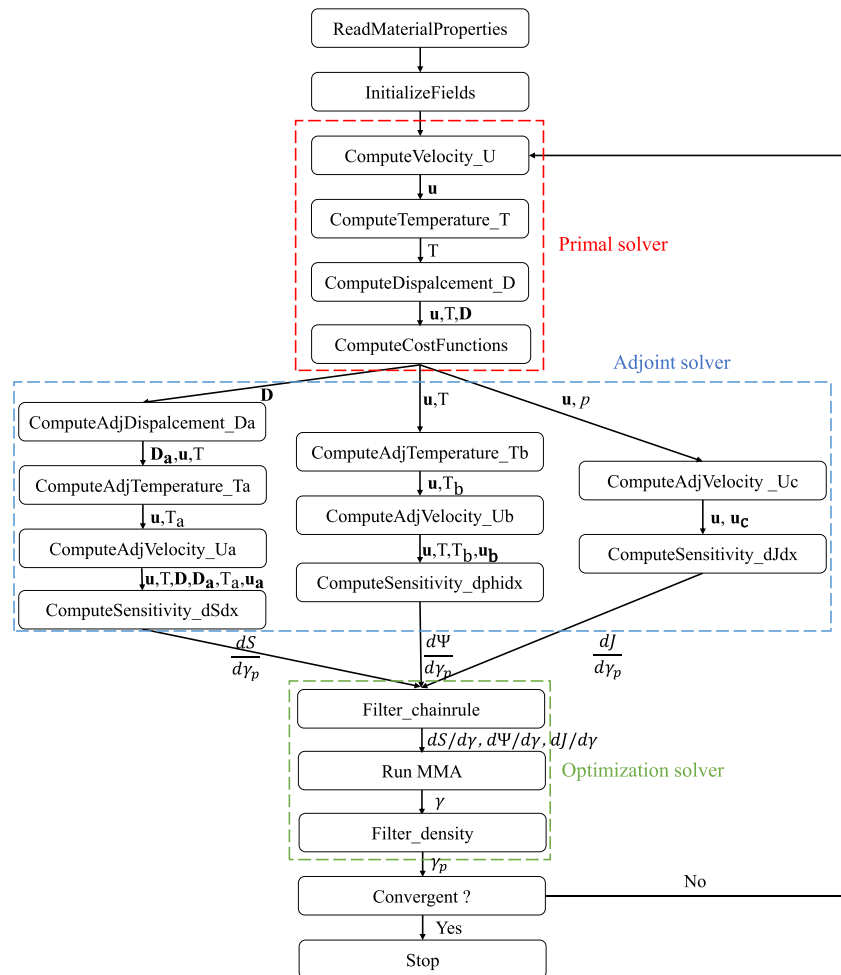
The density-based topology optimization of thermal-fluid-structural problems can be formulated as follows:

$$\begin{aligned} &\text{Find } \gamma \\ &\text{Minimize } \Psi \\ &\text{Subject to : } \int_{\Omega} \gamma d\Omega / |\Omega| \leq \beta && \text{Volume constraint} \\ &J < \bar{J} && \text{Power dissipated} \\ &S < \bar{S} && \text{Normal displacement} \\ &0 \leq \gamma \leq 1 && \text{Design variable bounds} \\ &\text{Equation (1)–(3)} && \text{Governing equations} \end{aligned} \quad (12)$$

where β is the volume fraction occupied by the fluid.

2.4 Implementation with OpenFOAM

The flowchart of the optimization process is presented in Fig. 3, where the blocks represent computational components, and the arrows indicate the data flow. Three main modules (primal solver, adjoint solver, and optimization solver) can be found in the flowchart. The primal and adjoint solvers are used to compute state and adjoint variables, respectively, and the optimization

Fig. 3 Flowchart of the optimization process

solver is composed of a filter and a gradient-based optimizer. The built-in solvers in OpenFOAM are used to solve weakly coupled multi-physics problems and their adjoint equations, see Table 1.

- N-S equations (1) are solved with simpleFoam.
- With the obtained velocity, scalarTransportFoam is adopted to solve the energy conservation equation (2).
- The obtained temperature field is then employed to estimate the thermal expansion in the linear elasticity equation (3) which is solved with solidDisplacementFoam.

The aforementioned solvers can also be modified to solve adjoint equations. More information about the adjoint method can be found in the Appendix A.

- SolidDisplacementFoam is also adopted to solve the adjoint linear elasticity equation.
- The convection term in the adjoint energy equation (31) is modified to $-\nabla \cdot [\rho c(-\mathbf{u})T_a]$ and solved with scalarTransportFoam.
- According to the work of Othmer (2008), a SIMPLE-type algorithm that can calculate the adjoint velocity and the

adjoint pressure in a segregated manner is used to solve adjoint N-S equations.

- For the optimization solver, a parallel implementation about the Method of Moving Asymptotes (MMA) in the PETSc framework written by Aage and Lazarov (2013) is linked with OpenFOAM.

In the present work, the Helmholtz PDE filter (Lazarov and Sigmund 2011), which is efficient for large-scale problems and parallel computing, is adopted to prevent numerical instabilities in topology optimization problems:

$$-r^2 \nabla^2 \gamma_p + \gamma_p = \gamma \quad (13)$$

$$\nabla \gamma_p \cdot \mathbf{n} = 0 \quad (14)$$

where γ_p and r denote the filtered density and the filter radius, respectively. The homogeneous Neumann boundary condition is imposed on the boundary of the design domain. (13) expresses that the filtered density is a solution of the Helmholtz equation, which can be solved in OpenFOAM by Listing 1.

Table 1 Solvers for primal and adjoint problems

Solvers	Primal problems	Adjoint problems
simpleFoam	N-S equations	Adjoint N-S equations
scalarTransportFoam	Energy conservation equation	Adjoint energy conservation equation
solidDisplacementFoam	Linear elasticity equation	Adjoint linear elasticity equation

```
1. solve(-fvm::laplacian(gammap)+fvm::Sp(1/r/r,gammap)-1/r/r*gamma);
```

Listing 1 Solving the Helmholtz equation (13) in OpenFOAM

where the namespace “fvm” returns matrix coefficients according to the operator, and the system of linear algebraic equations is solved by the function “solve.” The filtered density field (γ_p), which is represented by the

symbol “gammap” in Listing 2, is defined as an object of volScalarField. It contains the discrete values of γ_p and the boundary information, such as the zeroGradient boundary condition in line 13.

```
1. volScalarField gammap
2. (
3.     IOobject
4.     (
5.         "gammap",
6.         runTime.timeName(),
7.         mesh,
8.         IOobject::NO_READ,
9.         IOobject::AUTO_WRITE
10.    ),
11.    mesh,
12.    scalar(voluse), //initial value
13.    zeroGradientFvPatchScalarField::typeName //boundary condition(Eq.(14))
14. )
15. volScalarField gamma(gammap);
```

Listing 2 Definition of the pseudo-density field ‘gamma’ and the filtered field ‘gammap’

The sensitivity of the objective function (or constraint function) with respect to density γ is calculated by the chain rule:

$$\frac{dJ}{d\gamma} = \frac{dJ}{d\gamma_p} \frac{d\gamma_p}{d\gamma} \quad (15)$$

where $dJ/d\gamma_p$ can be calculated by the adjoint method. $dJ/d\gamma$ is also the solution of the Helmholtz equation:

where s denotes $dJ/d\gamma$ and s_p denotes $dJ/d\gamma_p$. The boundary condition of s is also the homogeneous Neumann type. (16) can be solved by the code in Listing 3 in which the definitions of s and s_p are omitted for simplicity. It should be noted that the sensitivity is scaled before solving (16) because if the magnitude of sensitivity is very small, the solver will stop after the tolerance is satisfied.

```
1. //Scale the sensitivities. gMax(x) returns the maximum value of x.
2. sp=sp/gMax(mag(sp.primitiveField()));
3.
4. //Compute the sensitivity of power dissipation with respect to x (Eq.(16)).
5. solve(-fvm::laplacian(s)+fvm::Sp(1/r/r,s)-1/r/r*sp);
```

Listing 3 Solving the Helmholtz equation (16) in OpenFOAM

$$-r^2 \nabla^2 s + s = s_p \quad (16)$$

The above user-defined solvers do not require any parallel-specific coding, and they can run with many parallel processes by using the command “mpirun” if the computation domain has been split into several subdomains. The use of OpenFOAM with its high-level math-like syntax allows users to focus on the physical and mathematical background of an optimization problem without being distracted by technical implementation issues. The source code of the solver and three tutorials on the standard minimum compliance problem and 2D/3D heat sinks are available online (github.com/MTopOpt/MTO).

3 Numerical verification

3.1 2D problem

The optimization of a 2D heat sink subjected to the constraints of power dissipation and fluid volume is first carried out. The computational domain consists of the main design area (colored with green) and fixed fluid inlet and outlet areas (colored with gray) (Fig. 4). The boundary conditions are as follows.

- Constant velocity and temperature at the inlet, and zero pressure and thermal isolation at the outlet.
- No-slip adiabatic boundary conditions are enforced on all other external walls.

- A uniform heat source Q is generated in the entire domain.

The inlet width is adopted as the characteristic length scale, and Reynolds number (Re) is about 100. The objective of the optimization is to minimize the mean temperature. Water and aluminum are selected as the fluid and the solid, respectively, and their physical properties are presented in Table 2.

In order to investigate the validity of the solver, a comparison is made with the finite difference (FD). The difference step is set to 0.001. When working with FD, governing equations must be fully converged because sensitivities are computed by subtracting two very close values. Therefore, the tolerance of the solver should be set to a small value (10^{-8} in the present study).

A uniform material distribution $\gamma=0.4$ is adopted as the initial structure, and the sensitivity maps of power dissipation and mean temperature are displayed in Fig. 5. Ten points were selected to perform FD (Fig. 5a), and the detailed values of sensitivities and relative errors ($|(f_{\text{adjoint}} - f_{\text{FD}})/f_{\text{FD}}|$) are presented in Table 3. The sensitivities of power dissipation calculated by the adjoint method manifested good agreement with FD values (less than 1%), and the relative errors of objective sensitivities were about 5%. Therefore, the parallel solver is capable of calculating correct sensitivities.

After investigating the validity of the sensitivity analysis, the topology optimization of the 2D heat sink is performed. First, the power dissipation of the straight cooling channel shown in Fig. 6 is calculated as the reference value

Fig. 4 Geometry and boundary conditions of the 2D model

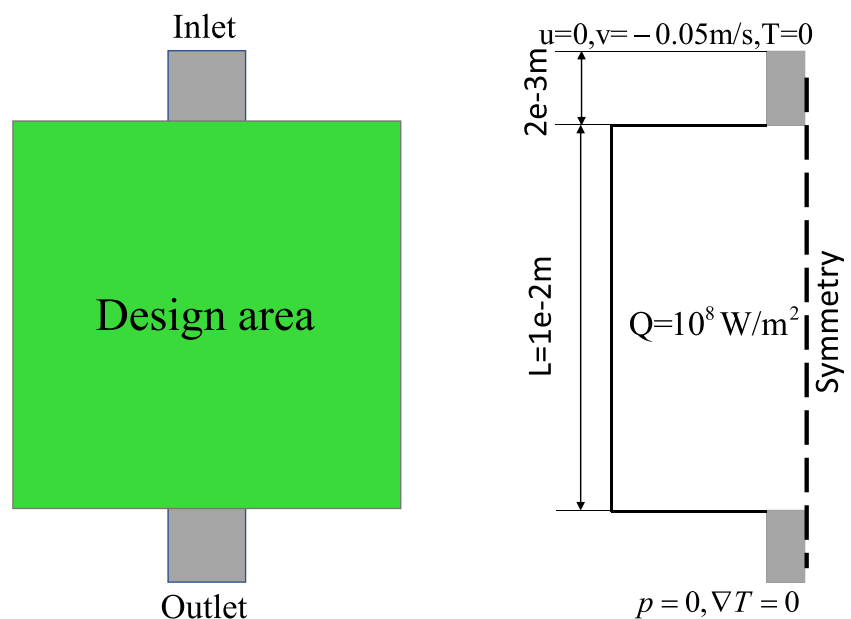


Table 2 Material properties of aluminum and water

Material	Water	Aluminum
k (W/(m · K))	0.61	237
c (J/(kg · K))	4180	880
ρ (kg/m ³)	1000	2700
η (N · s/m ²)	1e-3	–
E (N/m ²)	–	7e10
ν	–	0.33
α_{th} (1/K)	–	2e-5

($\underline{J}_1 = 1.58 \times 10^{-7}$). The effect of the power dissipation constraint on optimized designs was then investigated. The volume fraction occupied by the fluid was 40%. The optimized designs and corresponding temperature fields for different power dissipation constraints are displayed in Figs. 7, 8, and 9. The structural complexity and the cooling performance increased with the increase of the permissible power dissipation.

3.2 3D problem

In the current section, the topology optimization of 3D thermal-fluid-structural problems is studied. The geometry and boundary conditions of the 3D model are displayed in Fig. 10. In order to reduce the computational cost, the quarter model with 516,096 structured hexahedral cells is adopted as the simulation model. The optimization problem is solved on a computer with 20 Intel Xeon e5-2680v2 CPU cores. In order to achieve the best computing efficiency, first, the parallel speedup of one optimization loop with different numbers of subdomains is investigated. According to Fig. 11, all optimization problems are solved with 20 CPU cores. The volume fraction occupied by the fluid is 20%. The default settings of the built-in solver in

OpenFOAM including interpolation schemes and PDE solvers are adopted, and the tolerance of the solver is set to 10^{-6} (more detailed parameters can be found in [supplementary source files](#)). The computational times per iteration for thermal-fluid and thermal-fluid-structural problems are about 26 s and 40 s, respectively.

3.2.1 Darcy number for thermal-fluid problems

The parameter α_{max} has strong influences on the optimization results. Theoretically, α_{max} should be large enough to make the governing equations more consistent with the physical model. The dimensionless Darcy number, which describes the ratio between viscous and porous friction forces, can be expressed as

$$Da = \frac{\eta}{\alpha_{max} l^2} \quad (17)$$

where l is the characteristic length scale of the system. Olesen et al. (2006) studied 2D fluid flow problems and suggested that the Darcy number should be less than 10^{-5} to make the solid material impermeable. Dilgen et al. (2018b) adopted a lower Darcy number of 10^{-6} for 3D thermal-fluid problems. In the present experiment, the obtained optimization results are studied by setting the Darcy number as 10^{-5} , 10^{-6} , and 10^{-7} . The power dissipation of the 3D straight cooling channel shown in Fig. 12 is calculated as the reference value ($\underline{J}_2 = 9.4 \times 10^{-10}$). The permissible power dissipation is fourfold of the simplest design, $\bar{J} = 4\underline{J}_2$.

Selecting $l = 0.001$ m as the characteristic length scale, the maximum values of inverse permeability (α_{max}) are calculated as 10^5 , 10^6 , and 10^7 for $Da = 10^{-5}$, 10^{-6} , and 10^{-7} , respectively. It should be noted that if α_{max} is very high at the beginning of the optimization process, poor optimization results will be obtained to satisfy the power dissipation constraint. Hence, α_{max} is initially set as 5000 and increases gradually to its

Fig. 5 Sensitivity maps calculated by the adjoint solver with $\gamma = 0.4$: **a** sensitivities of power dissipation and **b** sensitivities of mean temperature

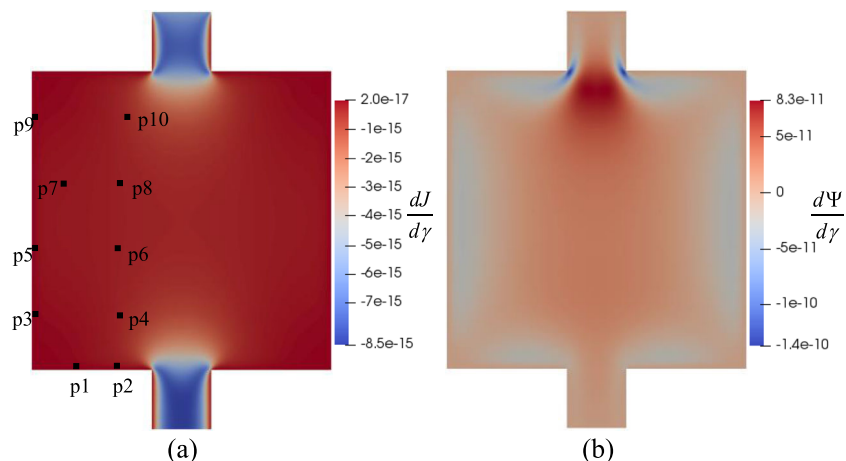


Table 3 Sensitivities and relative errors of the selected ten points

	Sensitivities of J			Sensitivities of Ψ		
	Difference	Continuous adjoint	Relative errors (%)	Difference	Continuous adjoint	Relative errors (%)
p1	-4.8203×10^{-18}	-4.8195×10^{-18}	0.0171	-1.8710×10^{-12}	-1.7442×10^{-12}	6.7777
p2	-3.7462×10^{-17}	-3.7597×10^{-17}	0.3607	-4.6612×10^{-12}	-4.9171×10^{-12}	5.4910
p3	-6.1644×10^{-18}	-6.1662×10^{-18}	0.0299	-3.0876×10^{-12}	-3.1567×10^{-12}	2.2403
p4	-3.0738×10^{-16}	-3.0811×10^{-16}	0.2400	1.0051×10^{-11}	9.8105×10^{-12}	2.3902
p5	-1.6464×10^{-17}	-1.6469×10^{-17}	0.0279	-3.9924×10^{-12}	-4.1944×10^{-12}	5.0586
p6	-2.6517×10^{-16}	-2.6472×10^{-16}	0.1684	1.5584×10^{-11}	1.5550×10^{-11}	0.2211
p7	-1.7051×10^{-16}	-1.7043×10^{-16}	0.0493	-2.0385×10^{-11}	-2.0431×10^{-11}	0.2284
p8	-2.6571×10^{-16}	-2.6509×10^{-16}	0.2310	1.7169×10^{-11}	1.6448×10^{-11}	4.2007
p9	-5.2372×10^{-18}	-5.2400×10^{-18}	0.0535	-2.6408×10^{-12}	-2.7903×10^{-12}	5.6641
p10	-3.9296×10^{-16}	-3.9213×10^{-16}	0.2110	1.7396×10^{-11}	1.6556×10^{-11}	4.8261

maximum value. Moreover, the parameters q_α and q_k grow from 0.005 to 0.02.

The optimal designs plotted by the Isovolume filter ranging from 0.5 to 1 in ParaView are displayed in Fig. 13, and their temperature distributions are compared in Fig. 14. All three designs consist of numerous cylindrical flow channels, and wall-like structures (marked by black circles in Fig. 13) are also observed in the cases of $Da = 10^{-5}$ and 10^{-6} . In these two cases, these wall-like structures divide the domain into several parts, and each part has a relatively uniform temperature

distribution. Theoretically, heat flows from the solid region to the fluid region by heat conduction and then is brought out of the heat sink by the fluid. However, the uniform temperature field implies a low heat conduction term. Therefore, the thermal flux transferred by conduction and convection in the solid region is calculated, and the parameter f is used to calculate the ratio of their magnitudes:

$$f = \frac{\left| \int_{\Gamma} k \nabla T \cdot \mathbf{n} d\Gamma \right|}{\left| \int_{\Gamma} \rho c T \mathbf{u} \cdot \mathbf{n} d\Gamma \right|} \quad (18)$$

The plane $z = -0.005$ is selected to calculate the values of f . The cross-sections of the investigated three designs and the values of f are plotted in Fig. 15. In an actual situation, convection cannot take place in solid areas; however, in numerical simulations, the convection effect should be significantly lower than heat conduction. In Fig. 15f, the values of f are about 3000 in solid areas, whereas in Fig. 15b and d, these values are about 15 and 100, respectively. Therefore, the Darcy number should be chosen carefully to avoid “wall-like” designs when dealing with thermal-fluid optimization problems.

In the following cases, the Darcy number is fixed at 10^{-7} . In order to compare the influences of stiffness constraints, a surface force of $\mathbf{F} = -30\text{MPa}$ is applied on the obtained optimal results (Fig. 13c), and the thermal expansion coefficient (α_{th}) is set to 0. The integral of the normal displacement is $S = 2.97 \times 10^{-16}$.

3.2.2 Thermal-fluid-structural problems without thermal expansion

It is assumed that temperature has no effect on structural displacement ($\alpha_{th} = 0$). An external load of $\mathbf{F} = -30\text{MPa}$ is

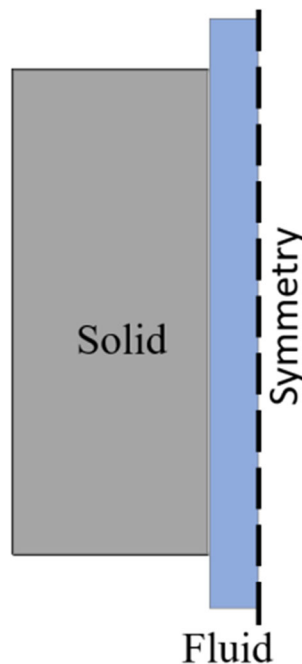


Fig. 6 Schematic diagram of the 2D straight cooling channel (blue and gray areas indicate the fluid and the solid, respectively)

Fig. 7 Optimal results of the 2D heat sink for $\bar{J} = 5J_1$: **a** pseudo-density field and **b** temperature field ($\Psi = 28.83$)

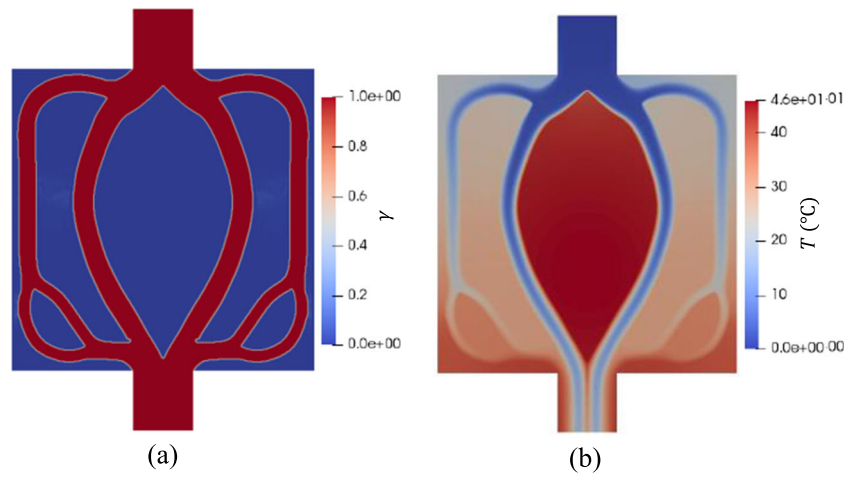


Fig. 8 Optimal results of the 2D heat sink for $\bar{J} = 10J_1$: **a** pseudo-density field and **b** temperature field ($\Psi = 19.05$)

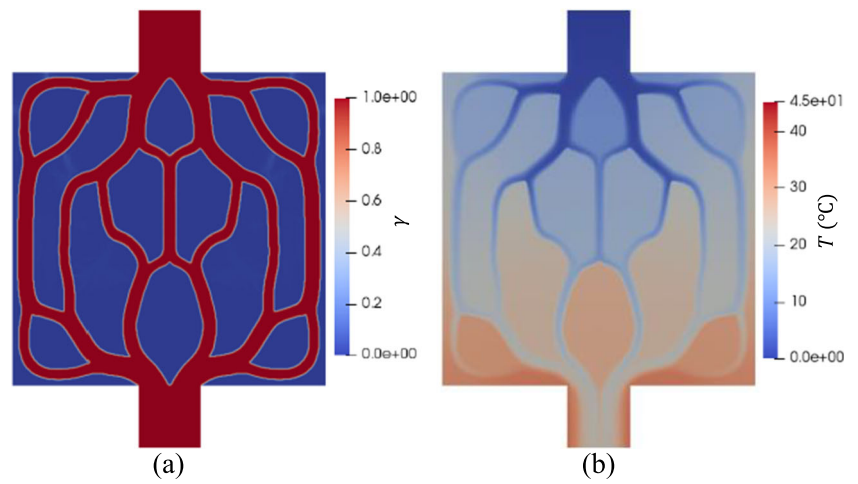
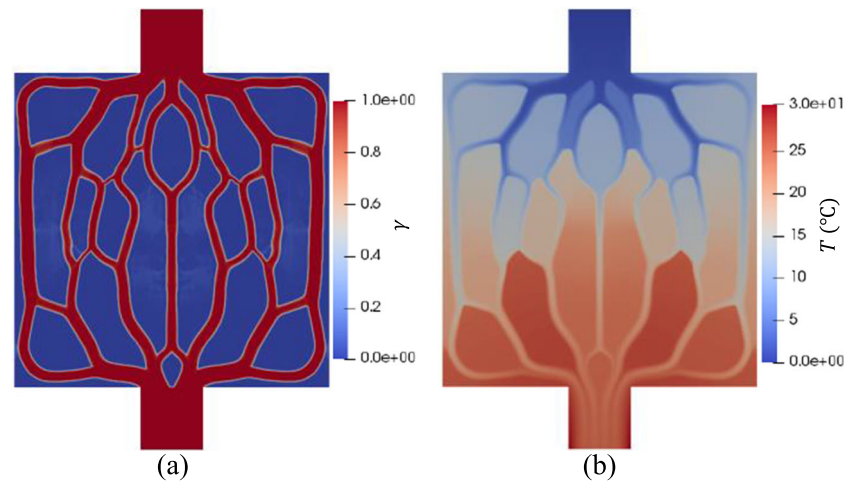


Fig. 9 Optimal results of the 2D heat sink for $\bar{J} = 15J_1$: **a** pseudo-density field and **b** temperature field ($\Psi = 16.96$)



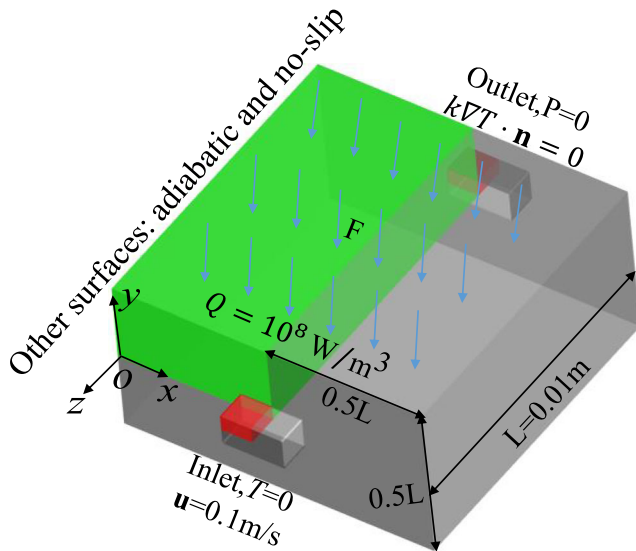


Fig. 10 Geometry and boundary conditions of the 3D model

applied on the top surface of the model. The permissible power dissipation is $\bar{J} = 4J_2$, and the displacement constraint for case B is $\bar{S} = 1.5 \times 10^{-16}$. Another stricter constraint $\bar{S} = 10^{-16}$ is applied to case C.

Figures 16 and 17 display the optimal pseudo-density fields of cases B and C, respectively, and the corresponding optimal temperature fields are presented in Figs. 18a and 19a, respectively. The mean temperatures of the optimal design for cases B and C are 51.28 °C and 53.31 °C, respectively. The surface displacement fields for cases B and C are displayed in Fig. 18 b and Fig. 19 b, respectively, and the cross-sections of the pseudo-density field for these two cases are presented in Fig. 20 and Fig. 21. The parameters for plotting these images are similar to those for case A.

Fig. 11 Speed up of the parallel optimization solver

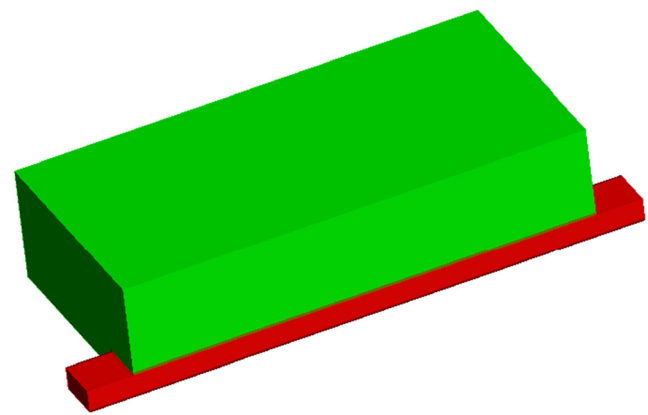
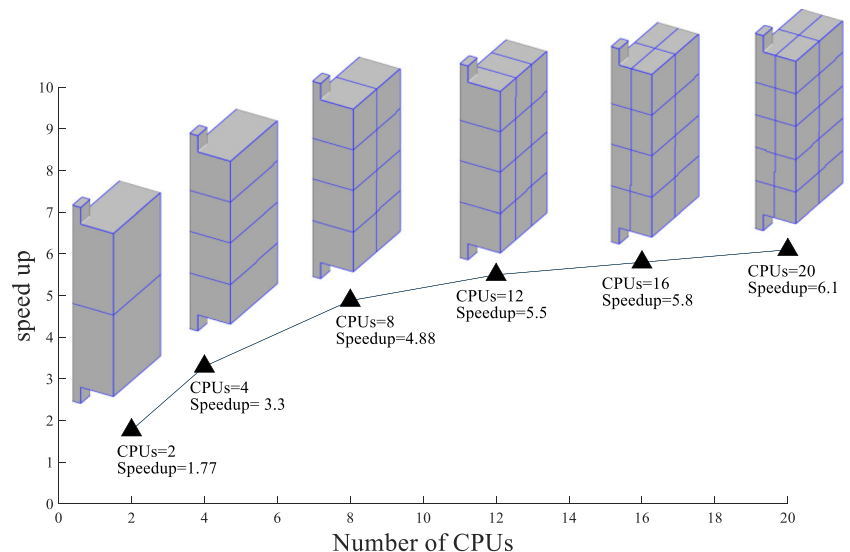


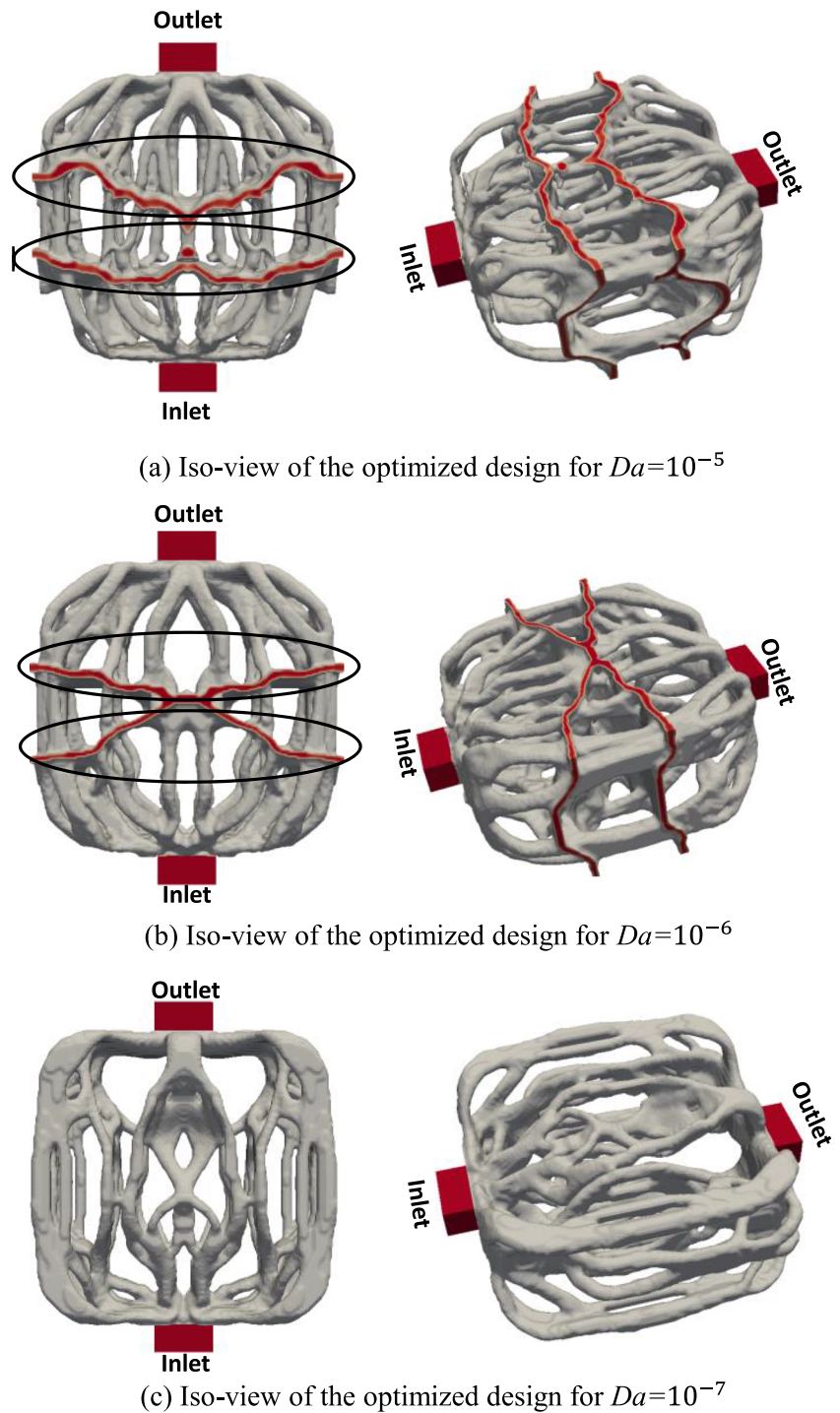
Fig. 12 Schematic diagram of the 3D straight cooling channel (red and green areas indicate the fluid and the solid, respectively)

In comparison to the thermal-fluid problem in case A, optimization results with the displacement constraint manifest quite different structural features. There are fewer branch channels, and the cross-sections of these channels have a larger aspect ratio. Continuous solid areas in the direction of the external force improve the structural stiffness and make the optimal structure more similar to a “2D extruded design.” Moreover, due to a stricter displacement constraint, these features are more obvious in case C than in case B (Figs. 16, 17, 22 and 23).

3.2.3 Thermal-fluid-structural problems with thermal expansion

The effects of thermal expansion are taken into consideration in cases D and E. The thermal expansion coefficient (α_{th}) is $\bar{S} = 2 \times 10^{-5}$. The values of external load and permissible

Fig. 13 Optimal results of the 3D thermal-fluid problem



power dissipation are similar to those in case B. As the deformation caused by the external force could be neutralized by thermal expansion, the constraint function S is set to a relatively small value for cases D ($\bar{S} = 10^{-17}$) and E ($\bar{S} = 5 \times 10^{-18}$).

Displacements caused by the thermal expansion and the external force are in opposite directions, leading to

more complex structural features in the optimal results of cases D and E than the other cases. Displacement caused by thermal expansion in the inlet area is smaller than that in the outlet area due to a lower temperature. It is discernible from Fig. 24a and Fig. 25a that there are fewer channels in comparison to case A, and some channels are similar to the “extruded design” in the z -direction

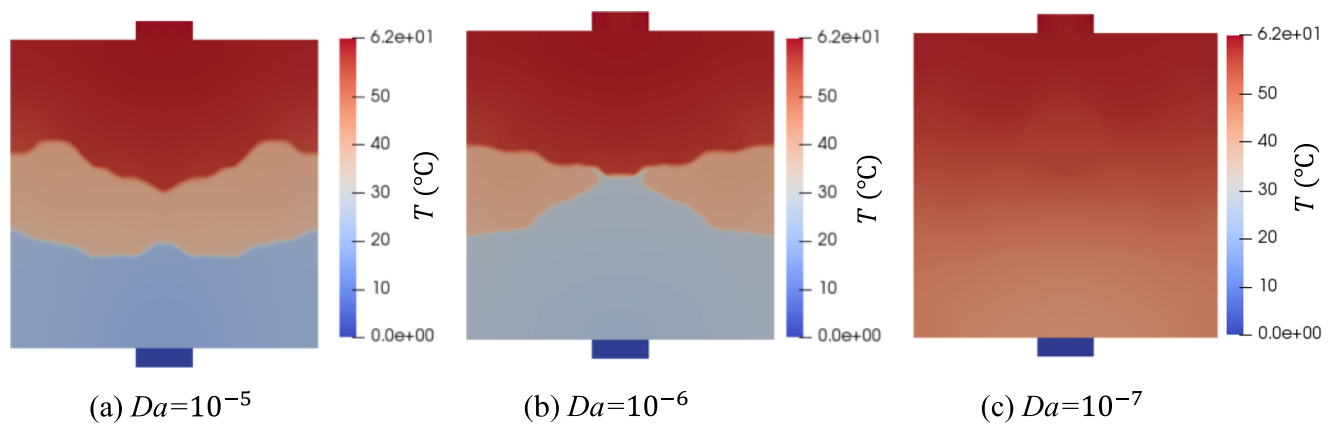


Fig. 14 Temperature fields of the obtained optimal results: **a** objective value = 39.90, $Da = 10^{-5}$; **b** objective value = 42.21, $Da = 10^{-6}$; and **c** objective value = 50.68, $Da = 10^{-7}$

(cases B and D), thus improving the structural stiffness to bear the external force. On the contrary, more fluid materials are used around the outlet area and the channels have a larger aspect ratio, improving the cooling performance and reducing the effect of thermal expansion. In addition, the more the area occupied by the fluid, the lower the structural stiffness; thus, displacements caused by thermal expansion could be neutralized by the external force (Fig.

26). Moreover, when the constraint is stricter, there exist more discontinuous fluid regions (marked by black circles in Fig. 27). As these isolated fluid regions have no cooling effect, therefore, it can be inferred that they existed only to reduce the stiffness.

The objective and constraint values of the above five optimal designs are compared in Table 3. Case A without any displacement constraint achieves the best cooling

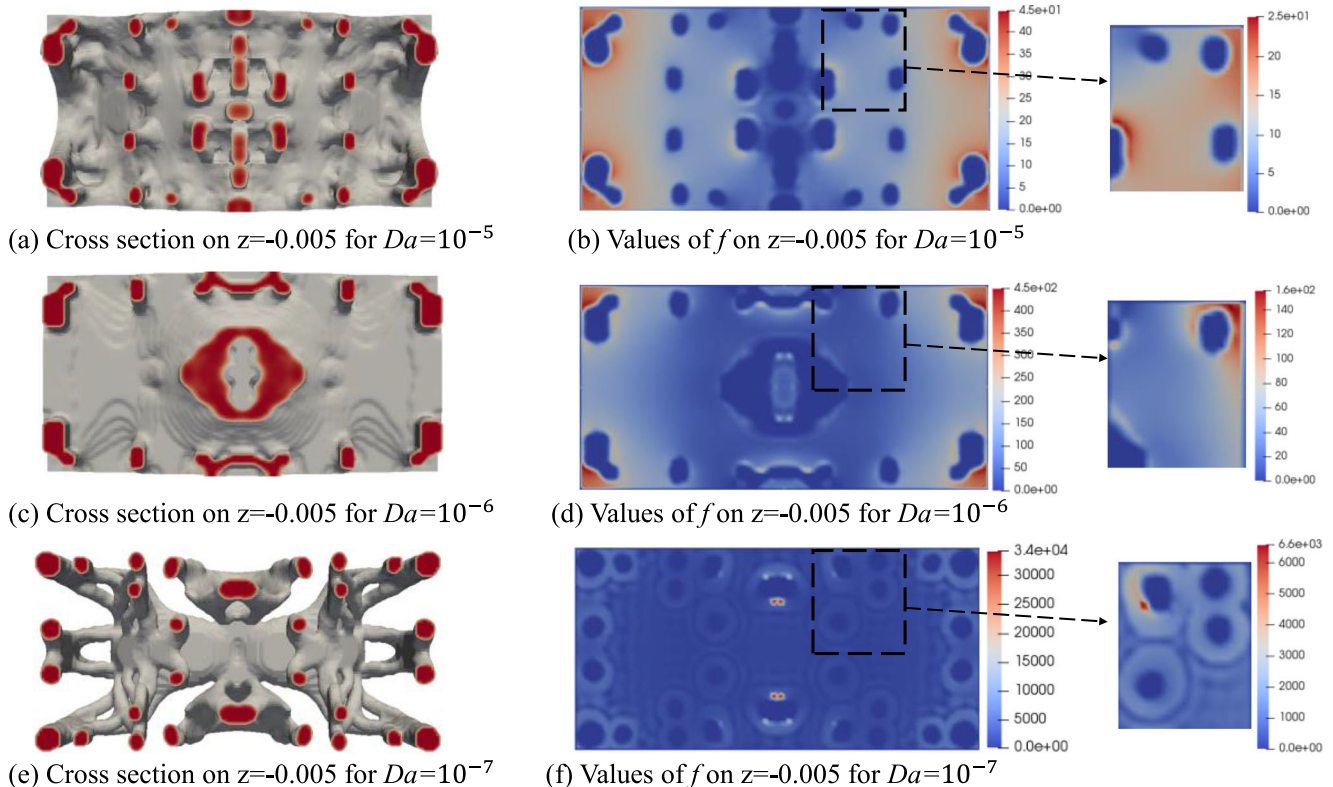


Fig. 15 Values of f on $z = -0.005$ for different Darcy numbers

Fig. 16 Optimal pseudo-density field of case B

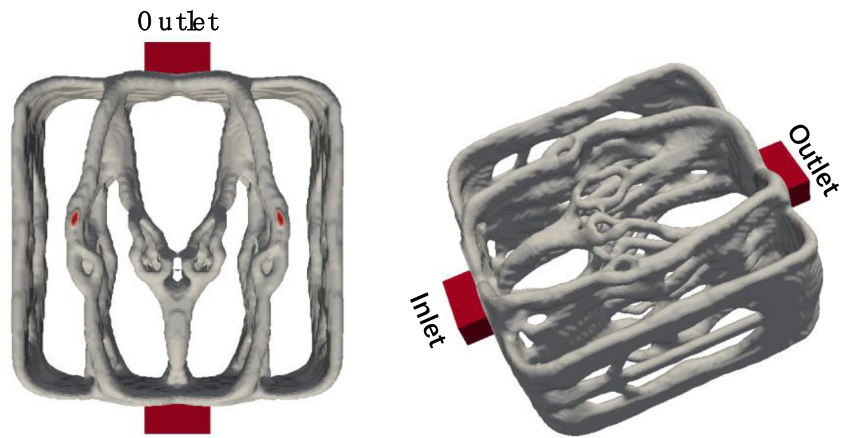


Fig. 17 Optimal pseudo-density and velocity fields of case C

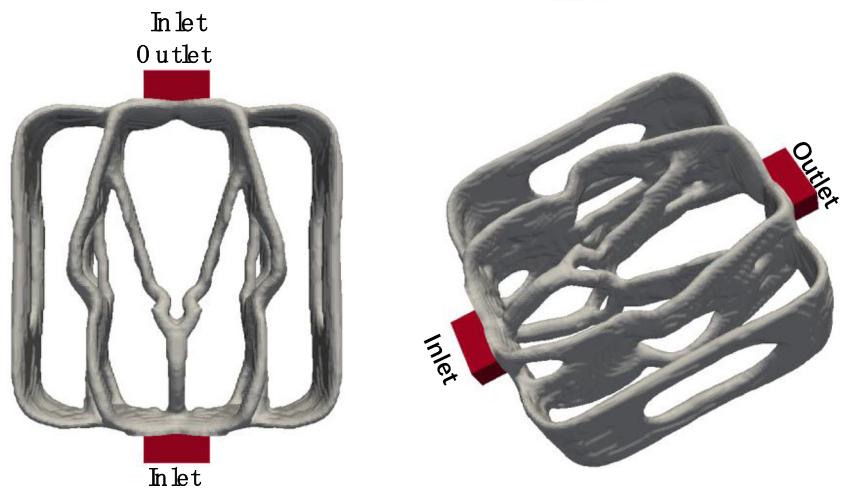


Fig. 18 Optimization results on the load-bearing surface of case B: **a** temperature field and **b** displacement (Y-component) field

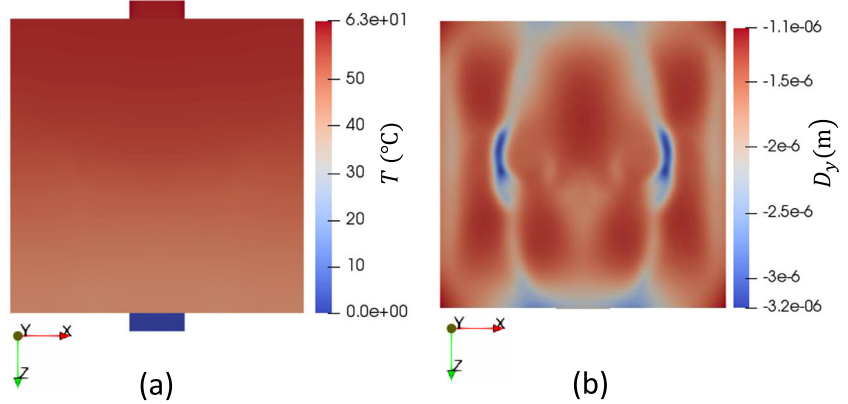
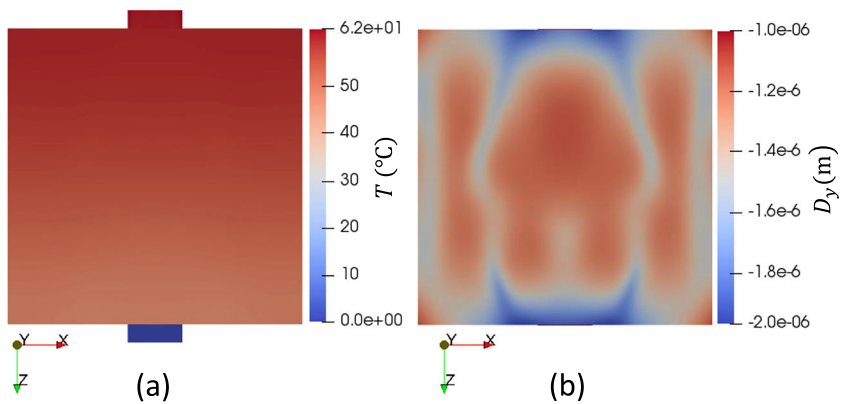


Fig. 19 Optimization results on the load-bearing surface of case C: **a** temperature field and **b** displacement (Y-component) field



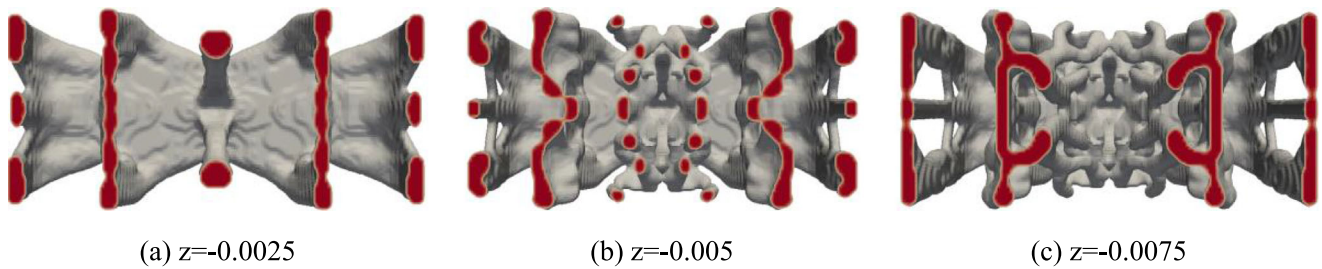


Fig. 20 Some cross-sections of the optimal result for case B

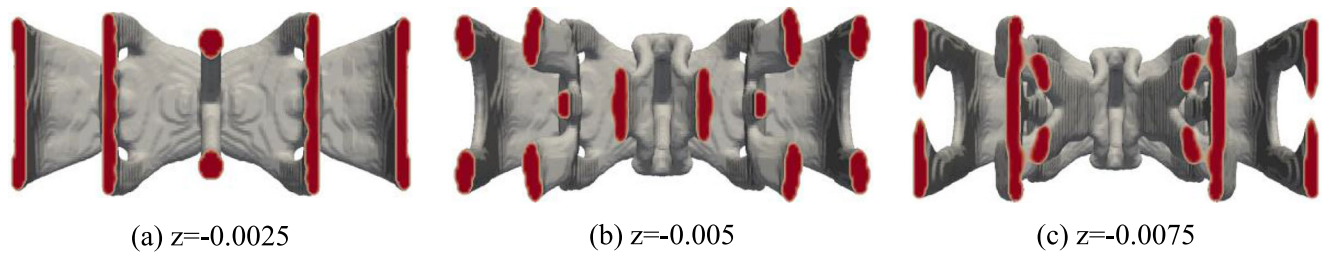


Fig. 21 Some cross-sections of the optimal result for case C

Fig. 22 Optimal pseudo-density and velocity fields of case D

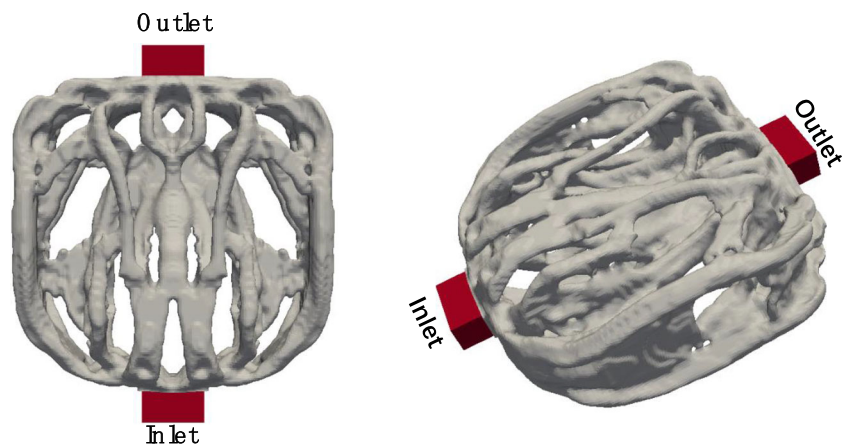
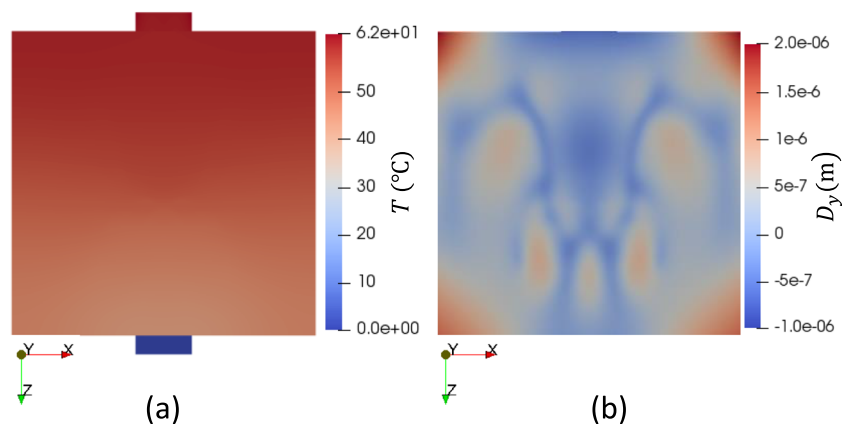


Fig. 23 Optimization results on the load-bearing surface of case D: **a** temperature field and **b** displacement (Y-component) field



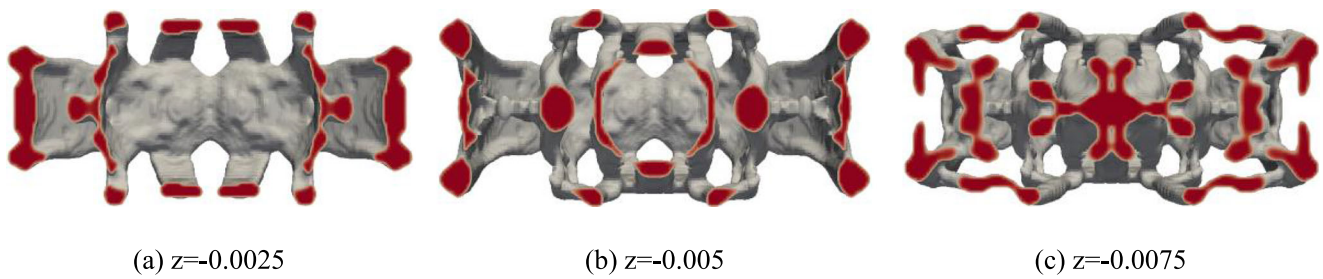


Fig. 24 Some cross-sections of the optimal result for case D. **a** $z=-0.0025$. **b** $z=-0.005$. **c** $z=-0.0075$

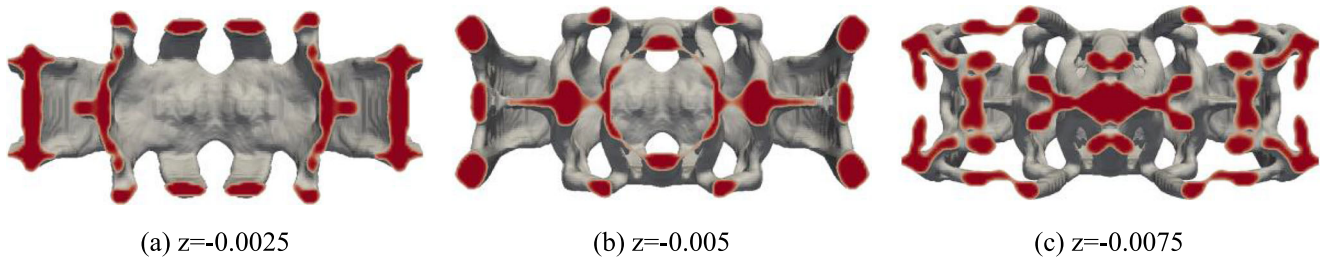


Fig. 25 Some cross-sections of the optimal result for case E

Fig. 26 Optimization results on the load-bearing surface of case E: **a** temperature field and **b** displacement (Y-component) field

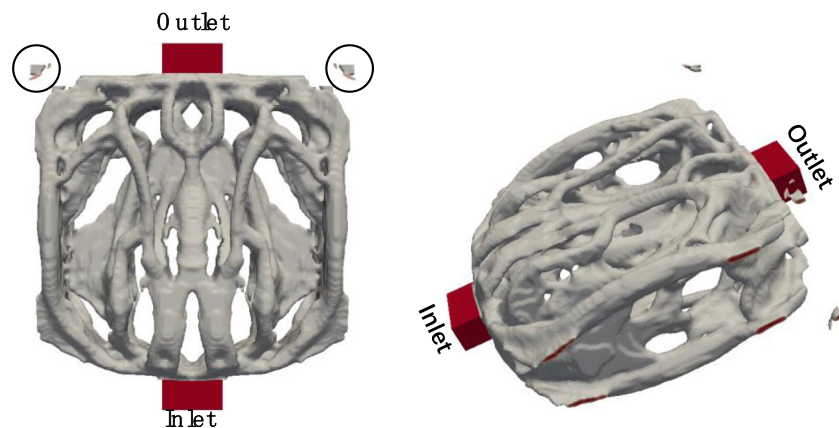
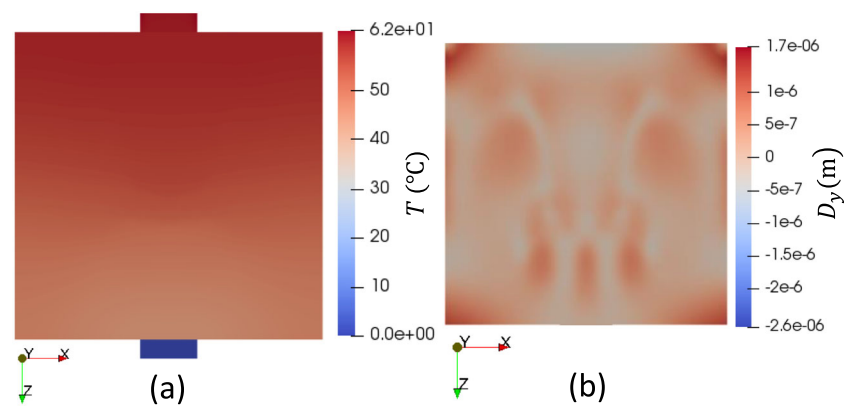


Fig. 27 Optimal pseudo-density and velocity fields of case E

Table 4 Objectives and constraints of the above five cases (upper limit of fluid volume = 0.2)

	Ψ	J	$\bar{J}$	S	$\bar{S}$	Fluid volume
Case A	50.68	$4J_2$	$4J_2$	—	—	0.195
Case B	51.28	$3.6J_2$	$4J_2$	1.5×10^{-16}	1.5×10^{-16}	0.185
Case C	53.31	$3.4J_2$	$4J_2$	10^{-16}	10^{-16}	0.159
Case D	51.50	$4J_2$	$4J_2$	10^{-17}	10^{-17}	0.190
Case E	51.74	$3.97J_2$	$4J_2$	5×10^{-18}	5×10^{-18}	0.192

performance. Moreover, for the studied thermal-fluid-structural problems with or without thermal expansion, a stricter constraint leads to a worse objective value. As there is no thermal expansion in cases B and C, the only way to satisfy the displacement constraint is to improve the stiffness; thus, the fluid volume dose not reach the threshold value (Table 4).

4 Conclusions

In the present study, the built-in solver in OpenFOAM is extended to solve multi-physics topology optimization problems. In order to take the advantages of object-oriented programming features in OpenFOAM, the continuous adjoint method is adopted to derive adjoint PDEs analytically, and then a custom parallel solver is built with ease. After verifying the accuracy of the solver on a 2D case, the effects of the Darcy number on thermal-fluid problems are discussed. Numerical tests reveal that although the Darcy number used in the fluid topology design could set the fluid velocity to near 0 in the solid region, it could not ensure the dominance of heat conduction over heat convection. The structural features of a 3D heat sink are also investigated. For the fluid-thermal optimization problem, channel cross-sections are either small with a low aspect ratio or large with a high aspect ratio. When displacement is considered a constraint, the optimal configuration became similar to a “2D extruded design,” which puts more solid regions in the direction of the external force to resist displacements. When displacements caused by thermal expansion are further considered, isolated fluid regions with no cooling ability exist and reduce the stiffness; thus, thermal displacements are neutralized by the external force.

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Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

Replication of results The source files and two tutorials are available as supplementary online materials for the convenience of researchers (github.com/MTopOpt/MTOT).

Appendix. Sensitivity analysis

The governing equations and boundary conditions for adjoint variables are derived analytically from the primal problem by adopting the continuous adjoint method, and subsequently, partial differential equations (PDEs) are discretized and solved. The detailed derivation of the studied thermal-fluid-structural topology optimization problems is presented below:

$$\text{minimize } \Psi \text{ subject to } \mathbf{R}(\mathbf{u}, p, T, \mathbf{D}, \gamma) = 0 \quad (19)$$

where $\mathbf{R} = (\mathbf{R}_1, R_2, R_3, \mathbf{R}_4)$ denotes state equations:

$$\begin{cases} R_1 = \rho(\mathbf{u} \cdot \nabla) \mathbf{u} - \eta \nabla \cdot (\nabla \mathbf{u} + \nabla \mathbf{u}^T) + \nabla p + \alpha_f \mathbf{u} \\ R_2 = -\nabla \cdot \mathbf{u} \\ R_3 = \rho c(\mathbf{u} \cdot \nabla T) - \nabla \cdot (k \nabla T) - Q \\ R_4 = \nabla \cdot \sigma \end{cases} \quad (20)$$

where $\sigma = \mu(\nabla \mathbf{D} + \nabla \mathbf{D}^T) + \lambda(\nabla \cdot \mathbf{D}) \mathbf{I} - \alpha_T(3\lambda + 2\mu)(T - T_{ref}) \mathbf{I}$.

Using the Lagrangian function L , the objective function can be rewritten as.

$$\begin{aligned} L &= \Psi + \langle \theta, \mathbf{R} \rangle \\ &= \Psi + \int_{\Omega} \mathbf{u}_a \cdot [\rho(\mathbf{u} \cdot \nabla) \mathbf{u} - \eta \nabla \cdot (\nabla \mathbf{u} + \nabla \mathbf{u}^T) + \nabla p + \alpha_f \mathbf{u}] d\Omega \\ &\quad + \int_{\Omega} p_a (-\nabla \cdot \mathbf{u}) d\Omega + \int_{\Omega} T_a [\rho c(\mathbf{u} \cdot \nabla T) - \nabla \cdot (k \nabla T) - Q] d\Omega \\ &\quad + \int_{\Omega} \mathbf{D}_a \cdot \{ \nabla \cdot [\mu(\nabla \mathbf{D} + \nabla \mathbf{D}^T) + \lambda(\nabla \cdot \mathbf{D}) \mathbf{I} - \alpha_T(3\lambda + 2\mu)(T - T_{ref}) \mathbf{I}] \} d\Omega \end{aligned} \quad (21)$$

where $\theta = (\mathbf{u}_a, p_a, T_a, \mathbf{D}_a)$ are Lagrangian multipliers (or adjoint variables), and $\langle \theta, \mathbf{R} \rangle = \int_{\Omega} (\theta \cdot \mathbf{R}) d\Omega$.

Now, employing integration by parts, (21) can be transformed into.

$$\begin{aligned} L &= \Psi + \int_{\Omega} [\rho(\mathbf{u} \cdot \nabla) \mathbf{u} \cdot \mathbf{u}_a + \eta(\nabla \mathbf{u} + \nabla \mathbf{u}^T) : \nabla \mathbf{u}_a - p \nabla \cdot \mathbf{u}_a + \alpha_f \mathbf{u} \cdot \mathbf{u}_a] d\Omega \\ &\quad + \int_{\Omega} p_a (-\nabla \cdot \mathbf{u}) d\Omega + \int_{\Omega} [\rho c(\mathbf{u} \cdot \nabla T) T_a + k \nabla T \cdot \nabla T_a - Q T_a] d\Omega \\ &\quad - \int_{\Omega} [\mu(\nabla \mathbf{D} + \nabla \mathbf{D}^T) + \lambda(\nabla \cdot \mathbf{D}) \mathbf{I} - \alpha_T(3\lambda + 2\mu)(T - T_{ref}) \mathbf{I}] : \nabla \mathbf{D}_a d\Omega \\ &\quad + \int_{\Gamma} \mathbf{u}_a \cdot [p \mathbf{I} - \eta(\nabla \mathbf{u} + \nabla \mathbf{u}^T)] \cdot \mathbf{n} d\Gamma - \int_{\Gamma} k T_a \nabla T \cdot \mathbf{n} d\Gamma \\ &\quad + \int_{\Gamma} \mathbf{D}_a \cdot [\mu(\nabla \mathbf{D} + \nabla \mathbf{D}^T) + \lambda(\nabla \cdot \mathbf{D}) \mathbf{I} - \alpha_T(3\lambda + 2\mu)(T - T_{ref}) \mathbf{I}] \cdot \mathbf{n} d\Gamma \end{aligned} \quad (22)$$

According to the Karush-Kuhn-Tucker conditions for PDE constrained optimization problems,

$$\begin{cases} L_u = \lim_{h \rightarrow 0} \frac{L(u + hv) - L(u)}{h} = 0 \\ L_p = \lim_{h \rightarrow 0} \frac{L(p + hq) - L(p)}{h} = 0 \\ L_T = \lim_{h \rightarrow 0} \frac{L(T + ht) - L(T)}{h} = 0 \\ L_D = \lim_{h \rightarrow 0} \frac{L(D + hd) - L(D)}{h} = 0 \end{cases} \quad (23)$$

where $\mathbf{v}$, q , t , $\mathbf{d}$ are the directions of the Gâteaux derivative.

The derivative of the augmented Lagrangian function with respect to γ is

$$\begin{aligned} \frac{dL}{d\gamma} &= \frac{\partial \Psi}{\partial \gamma} + \frac{\partial \Psi}{\partial u} \frac{\partial u}{\partial \gamma} + \frac{\partial \Psi}{\partial p} \frac{\partial p}{\partial \gamma} + \frac{\partial \Psi}{\partial T} \frac{\partial T}{\partial \gamma} + \frac{\partial \Psi}{\partial D} \frac{\partial D}{\partial \gamma} \\ &+ \frac{\partial \langle \theta, R \rangle}{\partial \gamma} + \frac{\partial \langle \theta, R \rangle}{\partial u} \frac{\partial u}{\partial \gamma} + \frac{\partial \langle \theta, R \rangle}{\partial p} \frac{\partial p}{\partial \gamma} + \frac{\partial \langle \theta, R \rangle}{\partial T} \frac{\partial T}{\partial \gamma} + \frac{\partial \langle \theta, R \rangle}{\partial D} \frac{\partial D}{\partial \gamma} \\ &= \left(\frac{\partial \Psi}{\partial u} + \frac{\partial \langle \theta, R \rangle}{\partial u} \right) \frac{\partial u}{\partial \gamma} + \left(\frac{\partial \Psi}{\partial p} + \frac{\partial \langle \theta, R \rangle}{\partial p} \right) \frac{\partial p}{\partial \gamma} \\ &+ \left(\frac{\partial \Psi}{\partial T} + \frac{\partial \langle \theta, R \rangle}{\partial T} \right) \frac{\partial T}{\partial \gamma} + \left(\frac{\partial \Psi}{\partial D} + \frac{\partial \langle \theta, R \rangle}{\partial D} \right) \frac{\partial D}{\partial \gamma} + \frac{\partial \Psi}{\partial \gamma} + \frac{\partial \langle \theta, R \rangle}{\partial \gamma} \end{aligned} \quad (24)$$

According to (23), (24) yields to

$$\frac{dL}{d\gamma} = \frac{\partial \Psi}{\partial \gamma} + \frac{\partial \langle \theta, R \rangle}{\partial \gamma}. \quad (25)$$

The adjoint variables must satisfy the following equations:

$$\begin{cases} L_u = \Psi_u + \int_{\Omega} [\rho(\mathbf{v} \cdot \nabla) \mathbf{u} \cdot \mathbf{u}_a + \rho(\mathbf{u} \cdot \nabla) \mathbf{v} \cdot \mathbf{u}_a + \eta(\nabla \mathbf{v} + \nabla \mathbf{v}^T) : \nabla \mathbf{u}_a + \alpha_f \mathbf{v} \cdot \mathbf{u}_a] d\Omega \\ \quad + \int_{\Omega} p_a (-\nabla \cdot \mathbf{v}) d\Omega + \int_{\Omega} \rho c (\mathbf{v} \cdot \nabla T) T_a d\Omega = 0 \\ L_p = \Psi_p - \int_{\Omega} q \nabla \cdot \mathbf{u}_a d\Omega + \int_{\Gamma_D^F} q u_a n d\Gamma = 0 \\ L_T = \Psi_T + \int_{\Omega} [\rho c (\mathbf{u} \cdot \nabla T) T_a + k \nabla T \cdot \nabla T_a + \alpha_T (3\lambda + 2\mu) I : \nabla D_a] d\Omega \\ \quad - \int_{\Gamma_D^T} [\alpha_T (3\lambda + 2\mu) I] D_a n d\Gamma = 0 \\ L_D = \Psi_D + \int_{\Omega} [-\mu(\nabla D + \nabla D^T) - \lambda(\nabla \cdot \mathbf{d}) I] : \nabla D_a d\Omega = 0 \end{cases} \quad (26)$$

The above equations can be reformulated into

$$\begin{cases} \Psi_u + \int_{\Omega} [\rho(\nabla \mathbf{u}) \cdot \mathbf{u}_a - \rho(\mathbf{u} \cdot \nabla) \mathbf{u}_a - \eta \nabla \cdot (\nabla \mathbf{u}_a + \nabla \mathbf{u}_a^T) + \alpha_f \mathbf{u}_a + \rho c T_a \nabla T] \cdot \mathbf{v} d\Omega \\ \quad + \int_{\Omega} \nabla p_a \cdot \mathbf{v} d\Omega + \int_{\Gamma_N^F} \{ \rho(\mathbf{u} \cdot \mathbf{n}) u_a + [-p_a I + \eta(\nabla \mathbf{u}_a + \nabla \mathbf{u}_a^T)] \cdot \mathbf{n} \} \cdot \mathbf{v} d\Gamma = 0 \\ \Psi_p - \int_{\Omega} q \nabla \cdot \mathbf{u}_a d\Omega + \int_{\Gamma_D^F} q u_a n d\Gamma = 0 \\ \Psi_T - \int_{\Omega} [\nabla \cdot (\rho c T_a \mathbf{u}) + k \nabla^2 T_a - \alpha_T (3\lambda + 2\mu) \nabla \cdot D_a] t d\Omega \\ \quad + \int_{\Gamma_N^T} [\rho c T_a \mathbf{u} + k \nabla T_a - \alpha_T (3\lambda + 2\mu) D_a] \cdot \mathbf{n} t d\Gamma = 0 \\ \Psi_D + \int_{\Omega} \nabla \cdot [\mu(\nabla D_a + \nabla D_a^T) + \lambda(\nabla \cdot D_a) I] \cdot \mathbf{d} d\Omega \\ \quad - \int_{\Gamma_N^S} [\mu(\nabla D_a + \nabla D_a^T) + \lambda(\nabla \cdot D_a) I] \cdot \mathbf{n} \cdot \mathbf{d} d\Gamma = 0 \end{cases} \quad (27)$$

Note that the integral on the Dirichlet boundary vanishes because the direction equals 0. Let consider a generic integral-

type objective function of the following form:

$$\begin{aligned} \Psi &= \int_{\Omega} A(\mathbf{u}, \nabla \mathbf{u}, p, T, \nabla T, \mathbf{D}, \nabla \mathbf{D}, \gamma) d\Omega \\ &\quad + \int_{\Gamma} B(\mathbf{u}, \nabla \mathbf{u}, p, T, \nabla T, \mathbf{D}, \nabla \mathbf{D}, \gamma) d\Gamma \end{aligned} \quad (28)$$

where A , B are objective functions defined in the domain and the boundary, respectively. The derivatives of the objective function with respect to state variables are

$$\begin{cases} \Psi_u = \int_{\Omega} \left(\frac{\partial A}{\partial u} - \nabla \cdot \frac{\partial A}{\partial \nabla u} \right) \cdot \mathbf{v} d\Omega + \int_{\Gamma_N^F} \left(\frac{\partial A}{\partial \nabla u} \cdot \mathbf{n} + \frac{\partial B}{\partial u} \right) \cdot \mathbf{v} d\Gamma \\ \Psi_p = \int_{\Omega} \frac{\partial A}{\partial p} \cdot q d\Omega + \int_{\Gamma_N^F} \frac{\partial B}{\partial p} \cdot q d\Gamma \\ \Psi_T = \int_{\Omega} \left(\frac{\partial A}{\partial T} - \nabla \cdot \frac{\partial A}{\partial \nabla T} \right) t d\Omega + \int_{\Gamma_N^T} \left(\frac{\partial A}{\partial \nabla T} \cdot \mathbf{n} + \frac{\partial B}{\partial T} \right) \cdot t d\Gamma \\ \Psi_D = \int_{\Omega} \left(\frac{\partial A}{\partial D} - \nabla \cdot \frac{\partial A}{\partial \nabla D} \right) \cdot \mathbf{d} d\Omega + \int_{\Gamma_N^S} \left(\frac{\partial A}{\partial \nabla D} \cdot \mathbf{n} + \frac{\partial B}{\partial D} \right) \cdot \mathbf{d} d\Gamma \end{cases} \quad (29)$$

Now substituting (29) into (27) and considering $\mathbf{v}$, q , t , $\mathbf{d}$ as arbitrary so that the integrand terms equal 0, the continuous adjoint equations and adjoint boundary conditions can be derived.

Adjoint linear elasticity equation:

$$\begin{cases} -\frac{\partial A}{\partial D} + \nabla \cdot \frac{\partial A}{\partial \nabla D} = \nabla \cdot [\mu(\nabla D_a + \nabla D_a^T) + \lambda(\nabla \cdot D_a) I] & \text{in } \Omega \\ D_a = 0 & \text{on } \Gamma_D^S \\ \frac{\partial A}{\partial \nabla D} \cdot \mathbf{n} + \frac{\partial B}{\partial D} = [\mu(\nabla D_a + \nabla D_a^T) + \lambda(\nabla \cdot D_a) I] \cdot \mathbf{n} & \text{on } \Gamma_N^S \end{cases} \quad (30)$$

Adjoint energy conservation equation:

$$\begin{cases} \frac{\partial A}{\partial T} - \nabla \cdot \frac{\partial A}{\partial \nabla T} = \nabla \cdot (\rho c T_a \mathbf{u}) + k \nabla^2 T_a - \alpha_T (3\lambda + 2\mu) \nabla \cdot D_a & \text{in } \Omega \\ T_a = 0 & \text{on } \Gamma_D^T \\ -\frac{\partial A}{\partial \nabla T} \cdot \mathbf{n} - \frac{\partial B}{\partial T} = [\rho c T_a \mathbf{u} + k \nabla T_a - \alpha_T (3\lambda + 2\mu) D_a] \cdot \mathbf{n} & \text{on } \Gamma_N^T \end{cases} \quad (31)$$

Adjoint N-S equations:

$$\begin{cases} \frac{\partial A}{\partial p} = \nabla \cdot \mathbf{u}_a & \text{in } \Omega \\ -\frac{\partial A}{\partial u} + \nabla \cdot \frac{\partial A}{\partial \nabla u} = \rho(\nabla \mathbf{u}) \cdot \mathbf{u}_a - \rho(\mathbf{u} \cdot \nabla) \mathbf{u}_a - \eta \nabla \cdot (\nabla \mathbf{u}_a + \nabla \mathbf{u}_a^T) + \alpha_f \mathbf{u}_a + \nabla p_a + \rho c T_a \nabla T \\ \quad - \frac{\partial B}{\partial p} = \mathbf{u}_a \cdot \mathbf{n} & \text{on } \Gamma_D^F \\ -\frac{\partial A}{\partial \nabla u} \cdot \mathbf{n} - \frac{\partial B}{\partial u} = \rho(\mathbf{u} \cdot \mathbf{n}) u_a + [-p_a I + \eta(\nabla \mathbf{u}_a + \nabla \mathbf{u}_a^T)] \cdot \mathbf{n} & \text{on } \Gamma_N^F \end{cases} \quad (32)$$

Except for the explicit objective (or constraint), such as volume constraint, the above adjoint system needs to be analyzed for every nonlinear objective (or constraint); however, not all adjoint PDEs need to be solved. For example, power dissipation J is independent from the solid displacement and heat transfer problem; thus, the adjoint variables $\mathbf{D}_a$ and T_a must be 0. From the mathematical point of view, the left-hand sides of (30) for J are 0 (Table 5), and it signifies that the

Table 5 Left-hand sides of adjoint equations for different objectives and constraints

	Ψ	S	J
$-\frac{\partial A}{\partial \mathbf{D}} + \nabla \cdot \frac{\partial A}{\partial \nabla \mathbf{D}}$	0	0	0
$\frac{\partial A}{\partial \nabla \mathbf{D}^T} \cdot \mathbf{n} + \frac{\partial B}{\partial \mathbf{D}}$	0	$2(\mathbf{D} \cdot \mathbf{n})\mathbf{n}$	0
$\frac{\partial A}{\partial T} - \nabla \cdot \frac{\partial A}{\partial \nabla T}$	Q	0	0
$-\frac{\partial A}{\partial \nabla T} \cdot \mathbf{n} - \frac{\partial B}{\partial T}$	0	0	0
$\frac{\partial A}{\partial p}$	0	0	0
$-\frac{\partial A}{\partial \mathbf{u}} + \nabla \cdot \frac{\partial A}{\partial \nabla \mathbf{u}}$	0	0	0
$-\frac{\partial B}{\partial p}$	0	0	$\mathbf{u} \cdot \mathbf{n}$
$-\frac{\partial A}{\partial \nabla \mathbf{u}^T} \cdot \mathbf{n} - \frac{\partial B}{\partial \mathbf{u}}$	0	0	$(p + 0.5\mathbf{u}^2)\mathbf{n} + (\mathbf{u} \cdot \mathbf{n})\mathbf{u}$

adjoint solid displacement problem bears no body or surface force; therefore, the adjoint displacement $\mathbf{D}_a$ is 0. Hence, only (32) is solved for the sensitivity of power dissipation, and (31) and (32) are solved for the sensitivity of mean temperature.

References

- Aage N, Lazarov BS (2013) Parallel framework for topology optimization using the method of moving asymptotes. *Struct Multidiscip Optim* 47:493–505. <https://doi.org/10.1007/s00158-012-0869-2>
- Aage N, Andreassen E, Lazarov BS, Sigmund O (2017) Giga-voxel computational morphogenesis for structural design. *Nature* 550:84–86. <https://doi.org/10.1038/nature23911>
- Alexandersen J, Sigmund O, Aage N (2016) Large scale three-dimensional topology optimisation of heat sinks cooled by natural convection. *Int J Heat Mass Tran* 100:876–891. <https://doi.org/10.1016/j.ijheatmasstransfer.2016.05.013>
- Au KM, Yu KM, Chiu WK (2011) Visibility-based conformal cooling channel generation for rapid tooling. *Comput Aided Design* 43:356–373. <https://doi.org/10.1016/j.cad.2011.01.001>
- Bendsoe MP, Kikuchi N (1988) Generating optimal topologies in structural design using a homogenization method. *Comput Methods Appl Mech Eng* 71:197–224. [https://doi.org/10.1016/0045-7825\(88\)90086-2](https://doi.org/10.1016/0045-7825(88)90086-2)
- Borrval T, Petersson J (2003) Topology optimization of fluids in stokes flow. *Int J Numer Methods Fluids* 41:77–107. <https://doi.org/10.1002/flid.426>
- Dang XP, Park HS (2011) Design of U-shape milled groove conformal cooling channels for plastic injection mold. *Int J Precision Eng Manuf* 12:73–84. <https://doi.org/10.1007/s12541-011-0009-8>
- Deng YB, Liu ZY, Zhang P, Wu YH, Korvink JG (2010) Optimization of no-moving part fluidic resistance microvalves with low Reynolds number. *Proc IEEE Micr Elect*:67–70. <https://doi.org/10.1109/MEMSYS.2010.5442565>
- Deng YB, Liu ZY, Zhang P, Liu YS, Wu YH (2011) Topology optimization of unsteady incompressible Navier-Stokes flows. *J Comput Phys* 230:6688–6708. <https://doi.org/10.1016/j.jcp.2011.05.004>
- Dilgen CB, Dilgen SB, Fuhrman DR, Sigmund O, Lazarov BS (2018a) Topology optimization of turbulent flows. *Comput Methods Appl Mech Eng* 331:363–393. <https://doi.org/10.1016/j.cma.2017.11.029>
- Dilgen SB, Dilgen CB, Fuhrman DR, Sigmund O, Lazarov BS (2018b) Density based topology optimization of turbulent flow heat transfer systems. *Struct Multidiscip Optim* 57:1905–1918. <https://doi.org/10.1007/s00158-018-1967-6>
- Duhring MB, Jensen JS, Sigmund O (2008) Acoustic design by topology optimization. *J Sound Vib* 317:557–575. <https://doi.org/10.1016/j.jsv.2008.03.042>
- Evgrafov A (2006) Topology optimization of slightly compressible fluids. *Zamm-Zeitschrift Fur Angewandte Mathematik Und Mechanik* 86:46–62. <https://doi.org/10.1002/zamm.200410223>
- Farrell PE, Ham DA, Funke SW, Rognes ME (2013) Automated derivation of the adjoint of high-level transient finite element programs. *SIAM J Sci Comput* 35:C369–C393. <https://doi.org/10.1137/120873558>
- Gersborg-Hansen A, Sigmund O, Haber RB (2005) Topology optimization of channel flow problems. *Struct Multidiscip Optim* 30:181–192. <https://doi.org/10.1007/s00158-004-0508-7>
- Gersborg-Hansen A, Bendsoe MP, Sigmund O (2006) Topology optimization of heat conduction problems using the finite volume method. *Struct Multidiscip Optim* 31:251–259. <https://doi.org/10.1007/s00158-005-0584-3>
- Giles MB, Pierce NA (2000) An introduction to the adjoint approach to design. *Flow Turbul. Combust* 65:393–415. <https://doi.org/10.1023/A:1011430410075>
- Guest JK, Prevost JH (2006) Topology optimization of creeping fluid flows using a Darcy-Stokes finite element. *Int J Numer Methods Eng* 66:461–484. <https://doi.org/10.1002/nme.1560>
- Kontoleonos E, Papoutsis-Kiachagias E, Zymaris A, Papadimitriou D, Giannakoglou K (2013) Adjoint-based constrained topology optimization for viscous flows, including heat transfer. *Eng Optimiz* 45:941–961. <https://doi.org/10.1080/0305215X.2012.717074>
- Kreissl S, Pingen G, Maute K (2011) Topology optimization for unsteady flow. *Int J Numer Methods Eng* 87:1229–1253. <https://doi.org/10.1002/nme.3151>
- Lazarov BS, Sigmund O (2011) Filters in topology optimization based on Helmholtz-type differential equations. *Int J Numer Methods Eng* 86:765–781. <https://doi.org/10.1002/nme.3072>
- Li CL (2001) A feature-based approach to injection mould cooling system design. *Computer Aided Design* 33:1073–1090. [https://doi.org/10.1016/S0010-4485\(00\)00144-5](https://doi.org/10.1016/S0010-4485(00)00144-5)
- Li CG, Li CL, Liu Y, Huang Y (2012) A new CC-space method to automate the layout design of injection mould cooling system. *Comput Aided Design* 44:811–823. <https://doi.org/10.1016/j.cad.2012.01.005>
- Li H, Ding X, Meng F, Jing D, MxJJJo H, Transfer M (2019a) Optimal design and thermal modelling for liquid-cooled heat sink based on multi-objective topology optimization: an experimental and numerical study. *Int J Therm Sci* 144:118638
- Li H, Ding XH, Jing DL, Xiong M, Meng FZ (2019b) Experimental and numerical investigation of liquid-cooled heat sinks designed by topology optimization. *Int J Thermal Sci* 146:UNSP 106065
- Lin JC (2002) Optimum cooling system design of a free-form injection mold using an abductive network. *J Mater Process Technol* 120:226–236. [https://doi.org/10.1016/S0010-4485\(00\)00144-5](https://doi.org/10.1016/S0010-4485(00)00144-5)
- Marta AC, Mader CA, Martins JRR, Van der Weide E, Alonso JJ (2007) A methodology for the development of discrete adjoint solvers using automatic differentiation tools. *Int J Comput Fluid D* 21:307–327. <https://doi.org/10.1080/10618560701678647>
- Nadarajah SK, Jameson A (2013) A comparison of the continuous and discrete adjoint approach to automatic aerodynamic optimization. *Can J Earth Sci*. <https://doi.org/10.2514/6.2000-667>
- Nomura T, Sato K, Taguchi K, Kashiwa T, Nishiwaki S (2007) Structural topology optimization for the design of broadband dielectric resonator antennas using the finite difference time domain technique. *Int*

- J Numer Methods Eng 71:1261–1296. <https://doi.org/10.1002/nme.1974>
- Nørgaard SA, Sagebaum M, Gauger NR, Lazarov BS (2017) Applications of automatic differentiation in topology optimization. *Struct Multidiscip Optim* 56:1135–1146. <https://doi.org/10.1007/s00158-017-1708-2>
- Olesen LH, Okkels F, Bruus H (2006) A high-level programming-language implementation of topology optimization applied to steady-state Navier-Stokes flow. *Int J Numer Methods Eng* 65:975–1001. <https://doi.org/10.1002/nme.1468>
- Othmer C (2008) A continuous adjoint formulation for the computation of topological and surface sensitivities of ducted flows. *Int J Numer Methods Fluids* 58:861–877. <https://doi.org/10.1002/fld.1770>
- Othmer C, de Villiers E, Weller H (2007) Implementation of a continuous adjoint for topology optimization of ducted flows. 18th AIAA Computational Fluid Dynamics Conference. <https://doi.org/10.2514/6.2007-3947>
- Papoutsis-Kiachagias EM, Giannakoglou KC (2014) Continuous adjoint methods for turbulent flows, applied to shape and topology optimization: industrial applications. *Arch Comput Methods Eng* 23:255–299. <https://doi.org/10.1007/s11831-014-9141-9>
- Park HS, Pham NH (2009) Design of conformal cooling channels for an automotive part. *Int J Automot Technol* 10:87–93. <https://doi.org/10.1007/s12239-009-0011-7>
- Peter JEV, Dwight RP (2010) Numerical sensitivity analysis for aerodynamic optimization: a survey of approaches. *Comput Fluids* 39:373–391. <https://doi.org/10.1016/j.compfluid.2009.09.013>
- Qiao H (2006) A systematic computer-aided approach to cooling system optimal design in plastic injection molding. *Int J Mech Sci* 48:430–439. <https://doi.org/10.1016/j.ijmecsci.2005.11.001>
- Sigmund O (1997) On the design of compliant mechanisms using topology optimization. *Mech Struct Mach* 25:493–524. <https://doi.org/10.1080/08905459708945415>
- Sigmund O (2001) A 99 line topology optimization code written in Matlab. *Struct Multidiscip Optim* 21:120–127. <https://doi.org/10.1007/s001580050176>
- Sigmund O, Hougaard K (2008) Geometric properties of optimal photonic crystals. *Phys Rev Lett* 100:ARTN 153904
- Wang XY, Li Z, Gu JF, Ruan SL, Shen CY, Wang XC (2016) Reducing service stress of the injection-molded polycarbonate window by optimizing mold construction and product structure. *Int J Adv Manuf Technol* 86:1691–1704. <https://doi.org/10.1007/s00170-015-8278-5>
- Wiker N, Klarbring A, Borrvall T (2007) Topology optimization of regions of Darcy and Stokes flow. *Int J Numer Methods Eng* 69:1374–1404. <https://doi.org/10.1002/nme.1811>
- Xu X, Sachs E, Allen S (2010) The design of conformal cooling channels in injection molding tooling. *Polym Eng Sci* 41:1265–1279. <https://doi.org/10.1002/pen.10827>
- Yaji K, Yamada T, Kubo S, Izui K, Nishiwaki S (2015) A topology optimization method for a coupled thermal–fluid problem using level set boundary expressions. *Int J Heat Mass Transf* 81:878–888. <https://doi.org/10.1016/j.ijheatmasstransfer.2014.11.005>
- Yaji K, Ogino M, Chen C, Fujita K (2018) Large-scale topology optimization incorporating local-in-time adjoint-based method for unsteady thermal–fluid problem. *Struct Multidiscip Optim* 58:817–822. <https://doi.org/10.1007/s00158-018-1922-6>
- Yoon GH (2010) Topological design of heat dissipating structure with forced convective heat transfer. *J Mech Sci Technol* 24:1225–1233. <https://doi.org/10.1007/s12206-010-0328-1>
- Yoon GH (2012) Topological layout design of electro-fluid-thermal-compliant actuator. *Comput Methods Appl Mech Eng* 209:28–44. <https://doi.org/10.1016/j.cma.2011.11.005>
- Yoon GH (2016) Topology optimization for turbulent flow with Spalart–Allmaras model. *Comput Methods Appl Mech Eng* 303:288–311. <https://doi.org/10.1016/j.cma.2016.01.014>
- Yu W, Yu KM, Wang CCL, Zhang Y (2011) Automatic design of conformal cooling circuits for rapid tooling. *Comput Aided Des* 43:1001–1010. <https://doi.org/10.1016/j.cad.2011.04.011>
- Yu M, Ruan S, Wang X, Li Z, Shen C (2019) Topology optimization of thermal–fluid problem using the MMC-based approach. *Struct Multidiscip Optim* 60:151–165. <https://doi.org/10.1007/s00158-019-02206-w>
- Zhao X, Zhou M, Liu Y, Ding M, Hu P, Zhu P (2019) Topology optimization of channel cooling structures considering thermomechanical behavior. *Struct Multidiscip Optim* 1–20. <https://doi.org/10.1007/s00158-018-2087-z>

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